

BFS Traversal

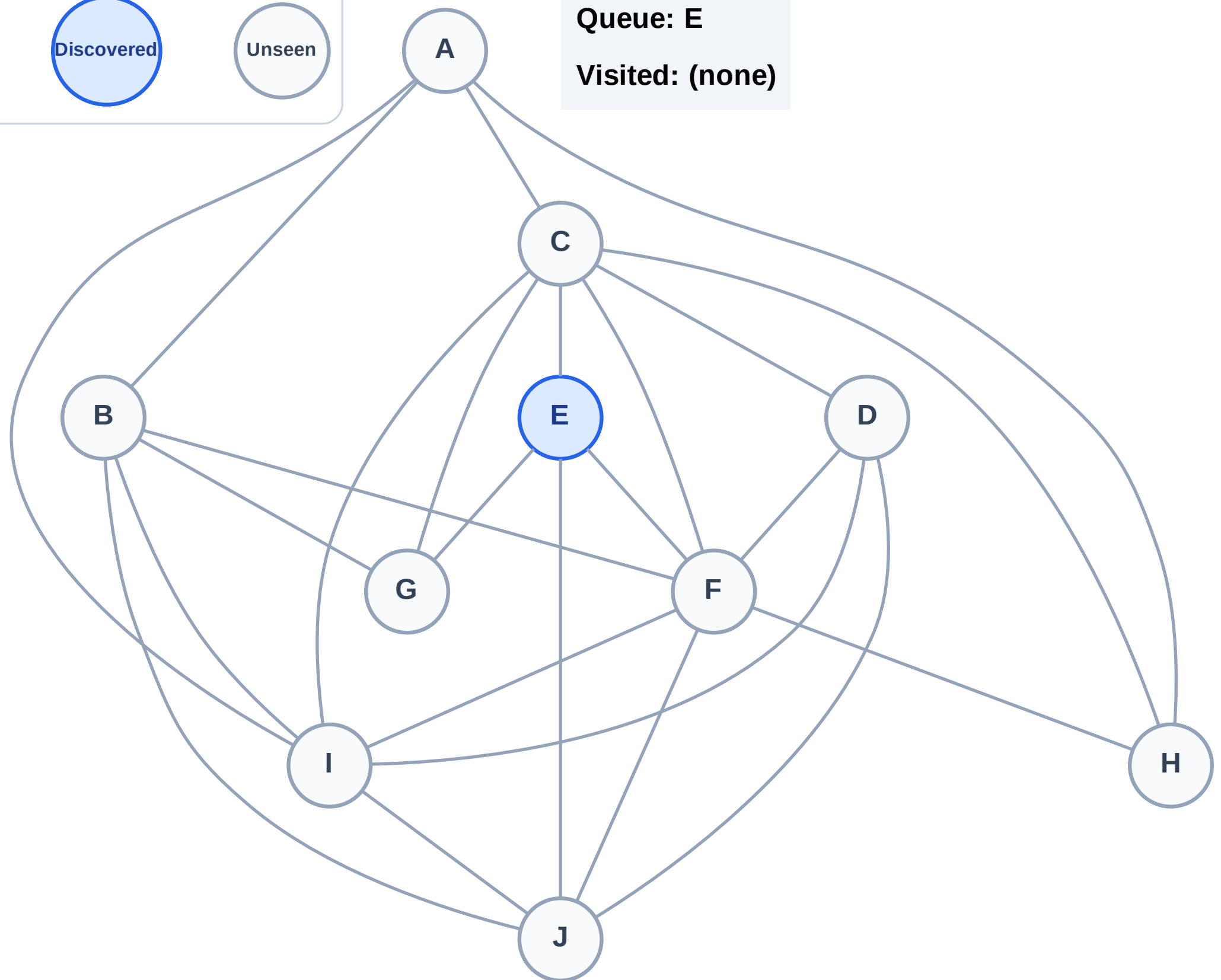
Step 1: Start at E

Legend



Queue: E

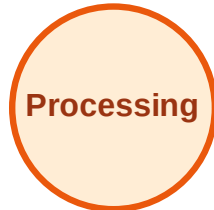
Visited: (none)



BFS Traversal

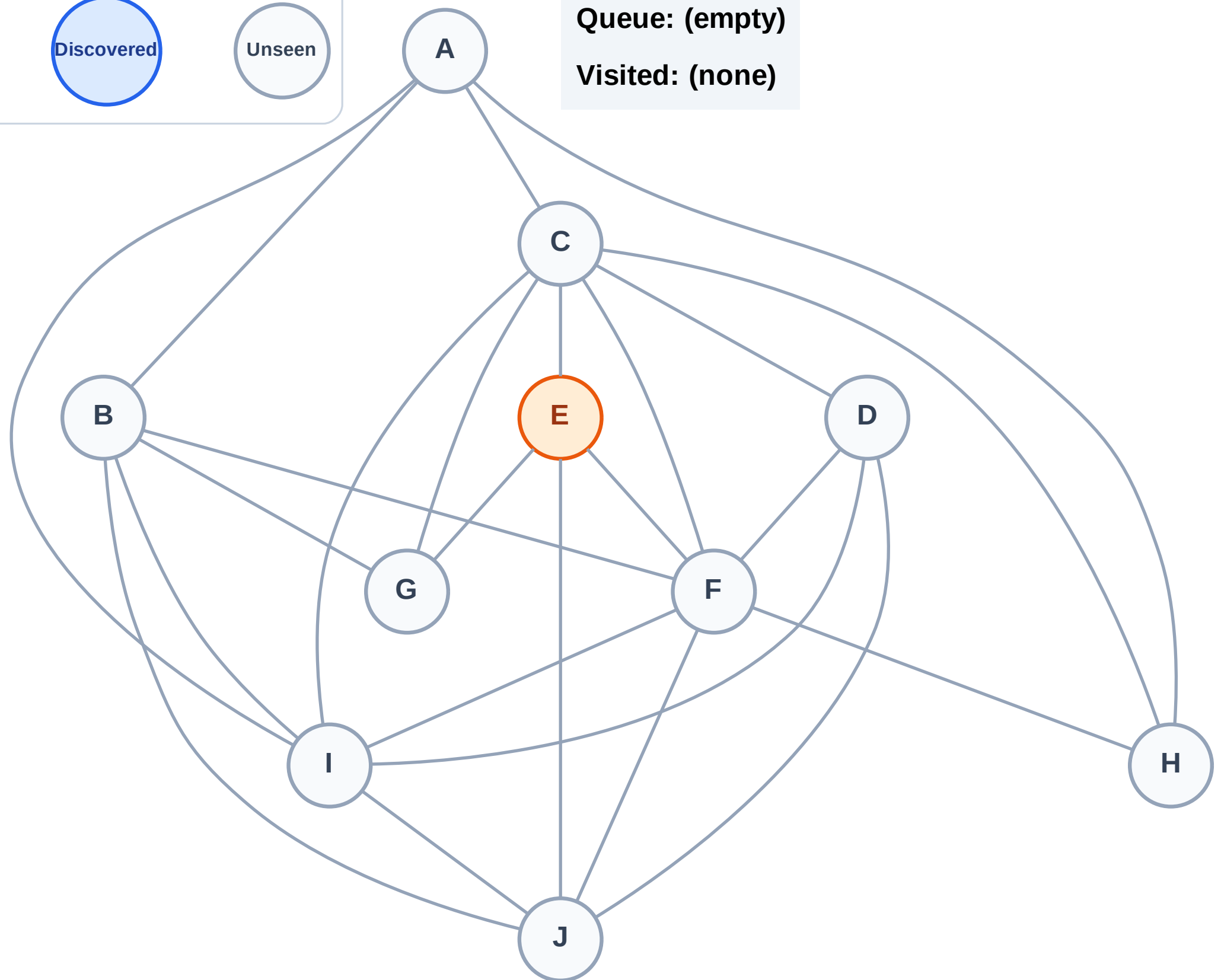
Step 2: Process node E

Legend



Queue: (empty)

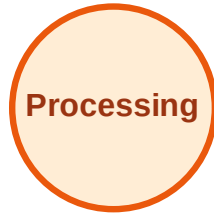
Visited: (none)



BFS Traversal

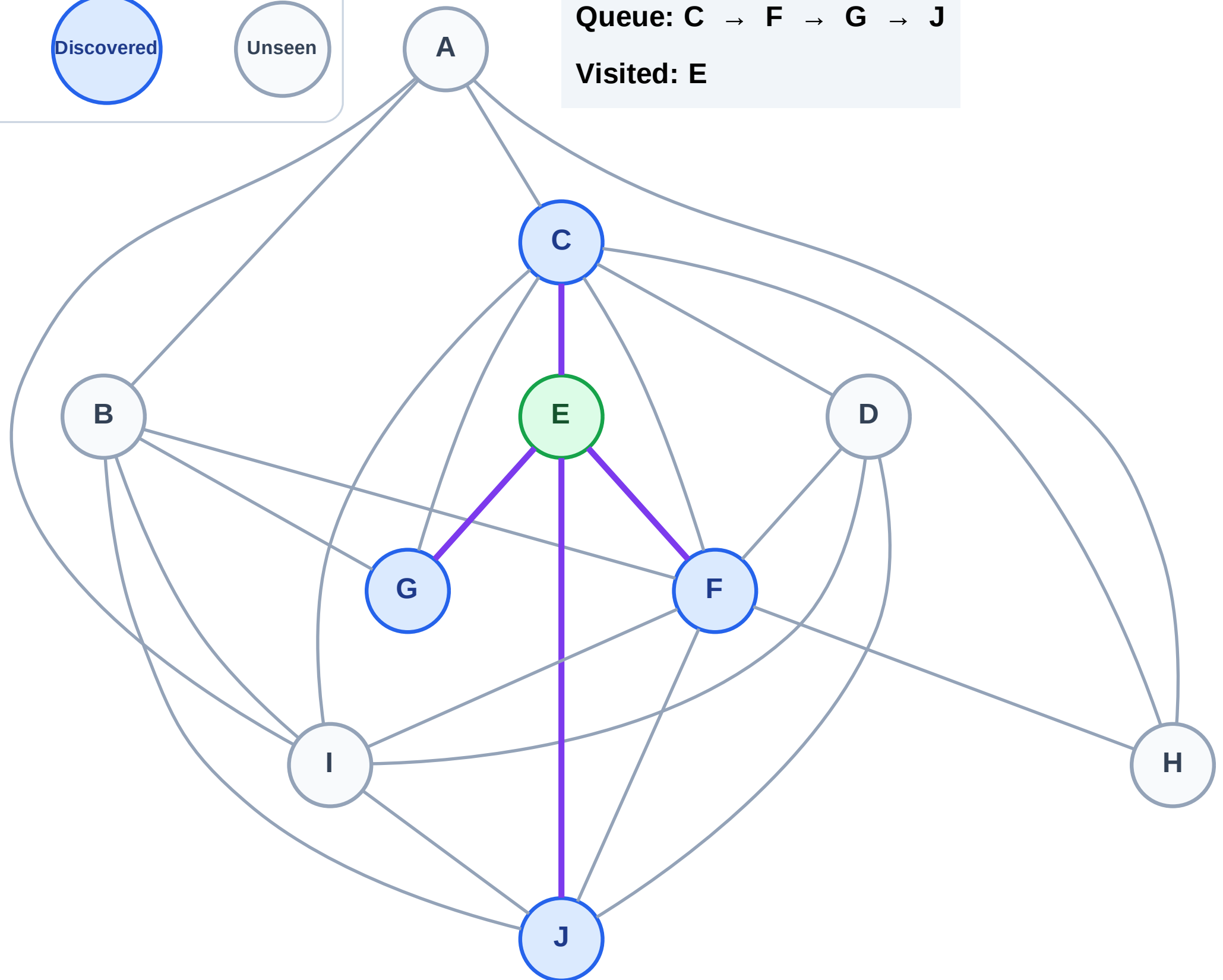
Step 3: Discover C, F, G, J from E

Legend



Queue: C → F → G → J

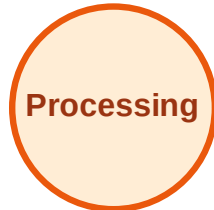
Visited: E



BFS Traversal

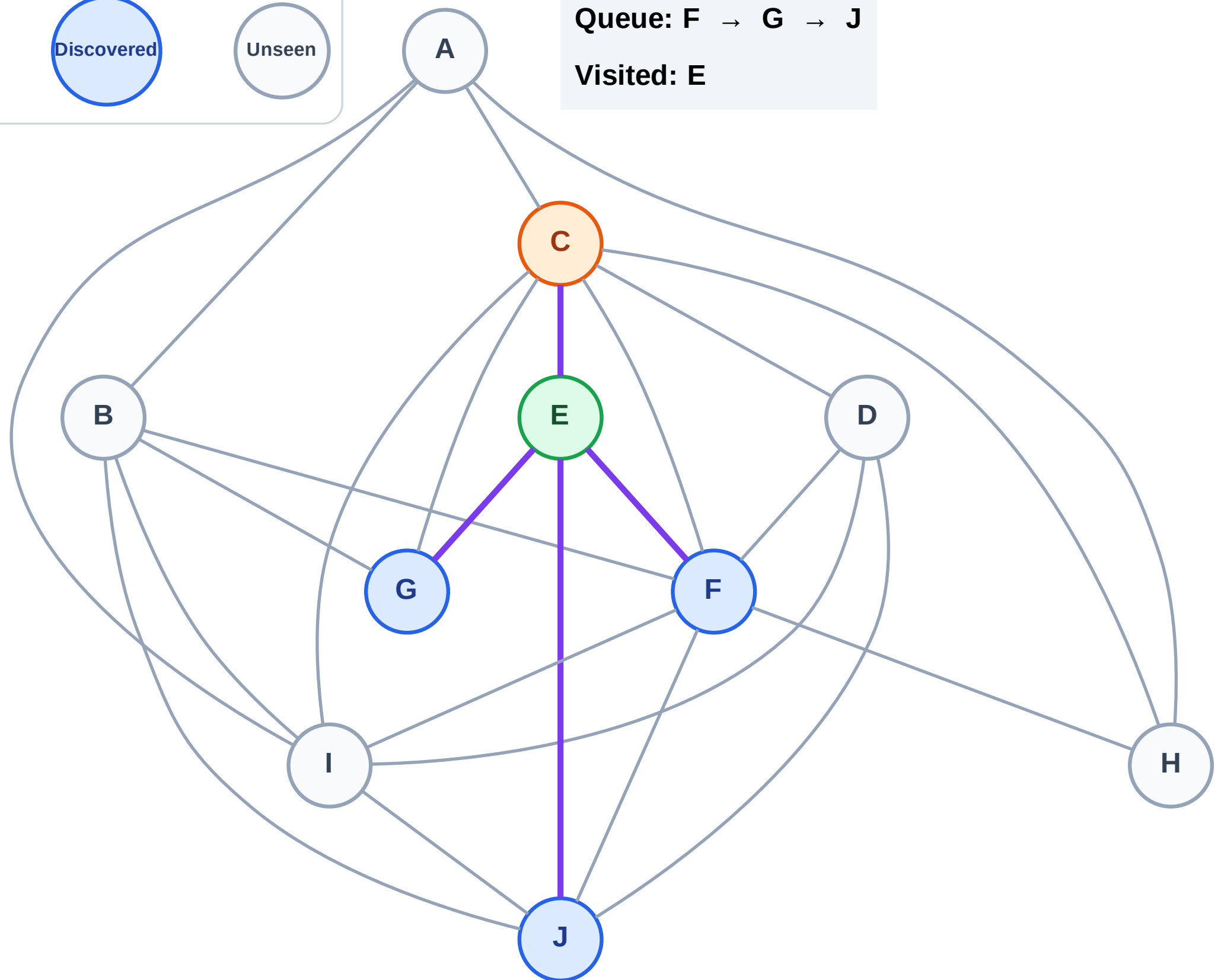
Step 4: Process node C

Legend



Queue: F → G → J

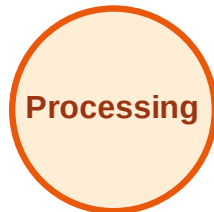
Visited: E



BFS Traversal

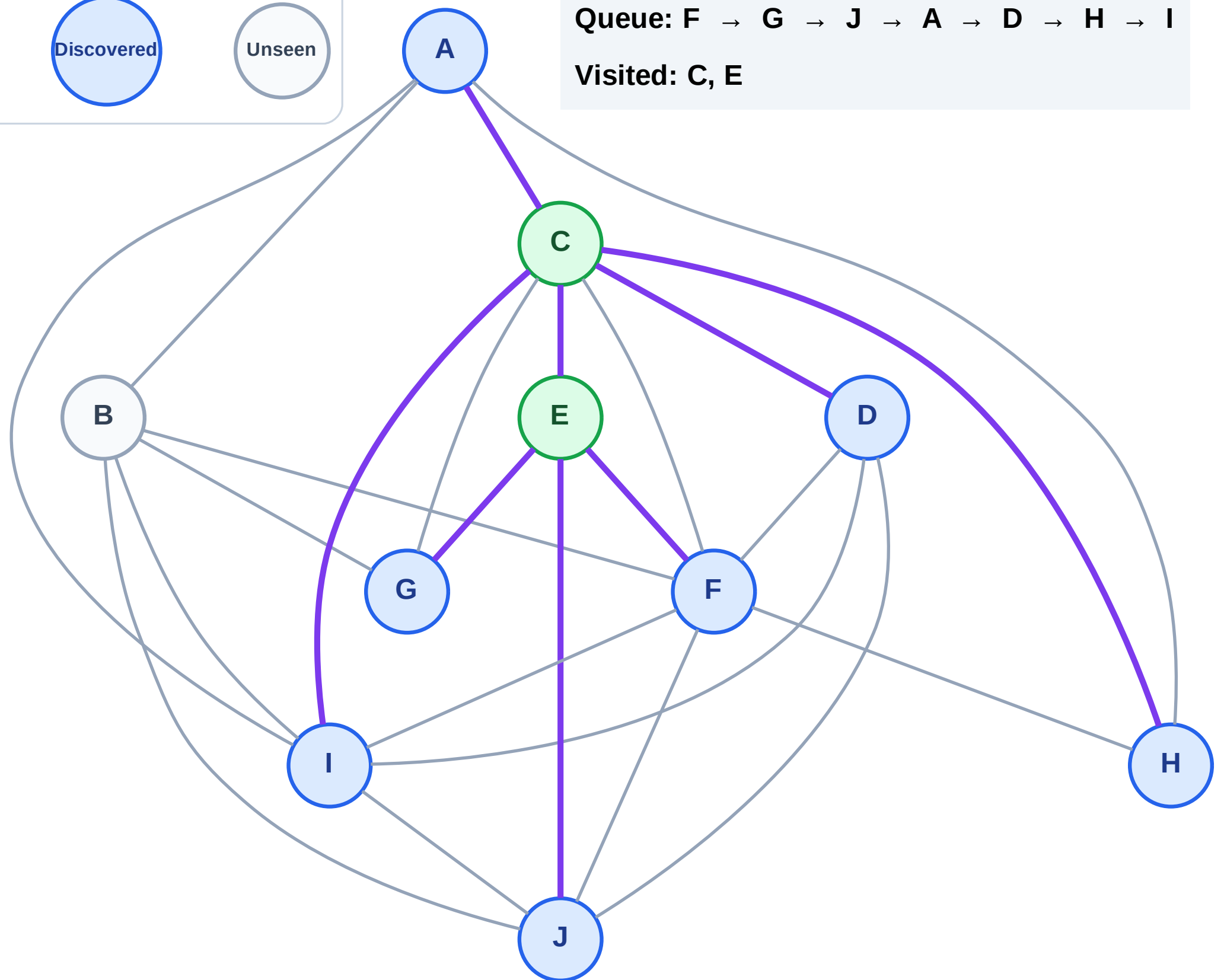
Step 5: Discover A, D, H, I from C

Legend



Queue: F → G → J → A → D → H → I

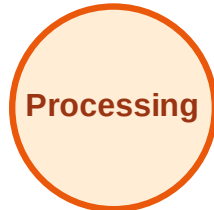
Visited: C, E



BFS Traversal

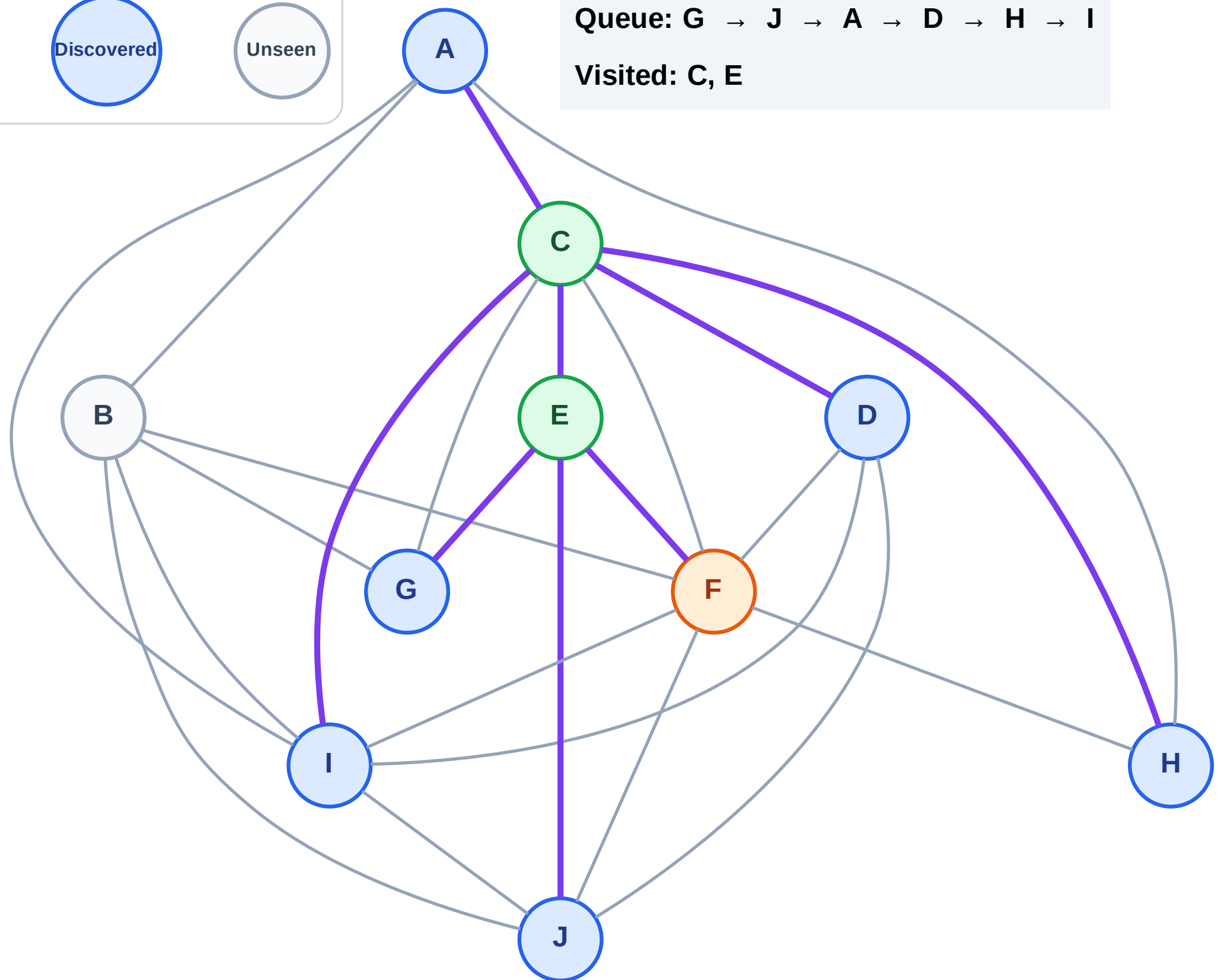
Step 6: Process node F

Legend



Queue: G → J → A → D → H → I

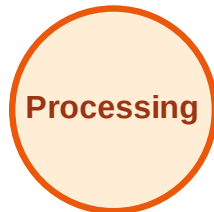
Visited: C, E



BFS Traversal

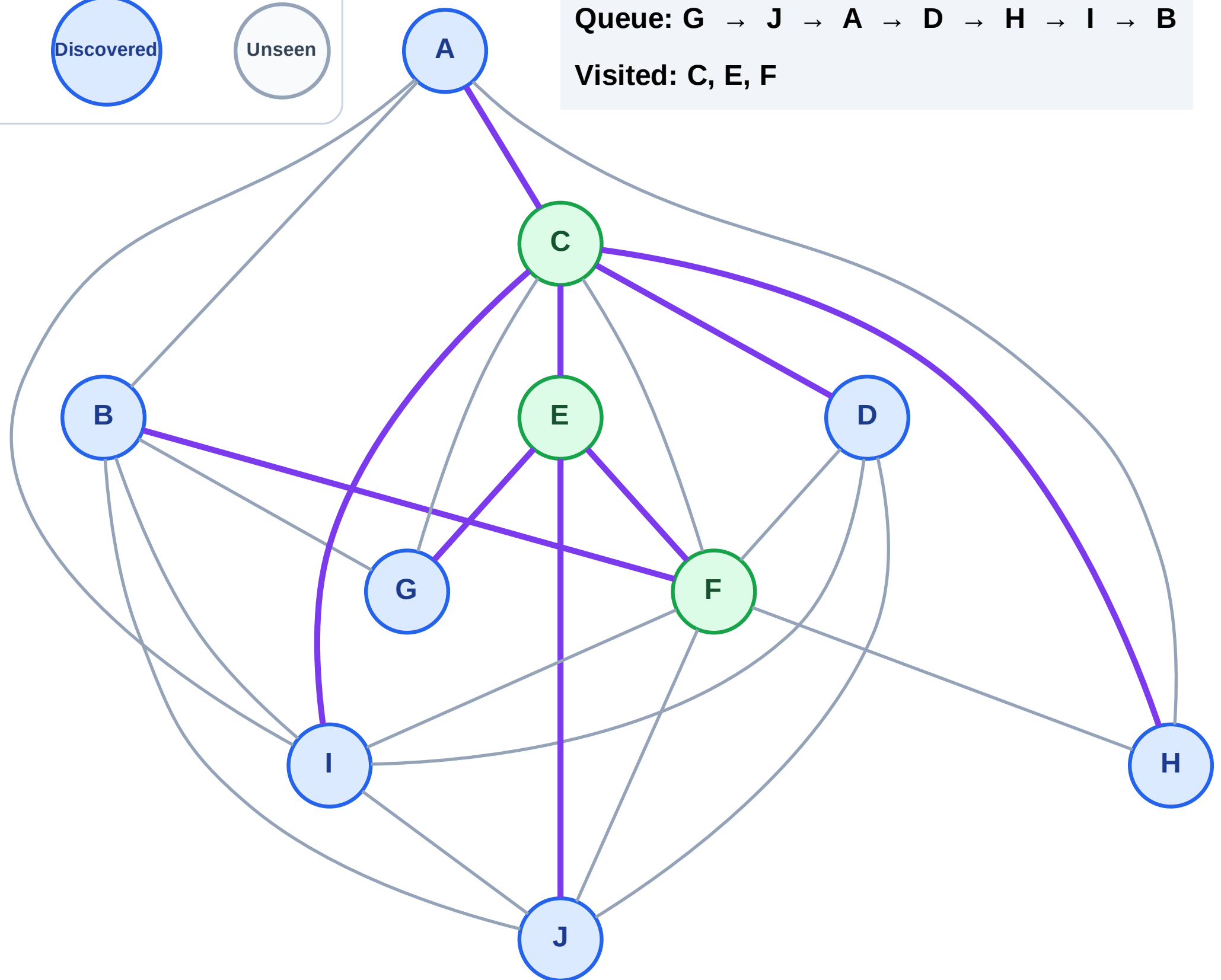
Step 7: Discover B from F

Legend



Queue: G → J → A → D → H → I → B

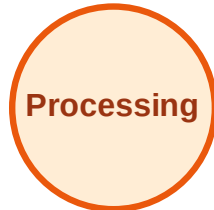
Visited: C, E, F



BFS Traversal

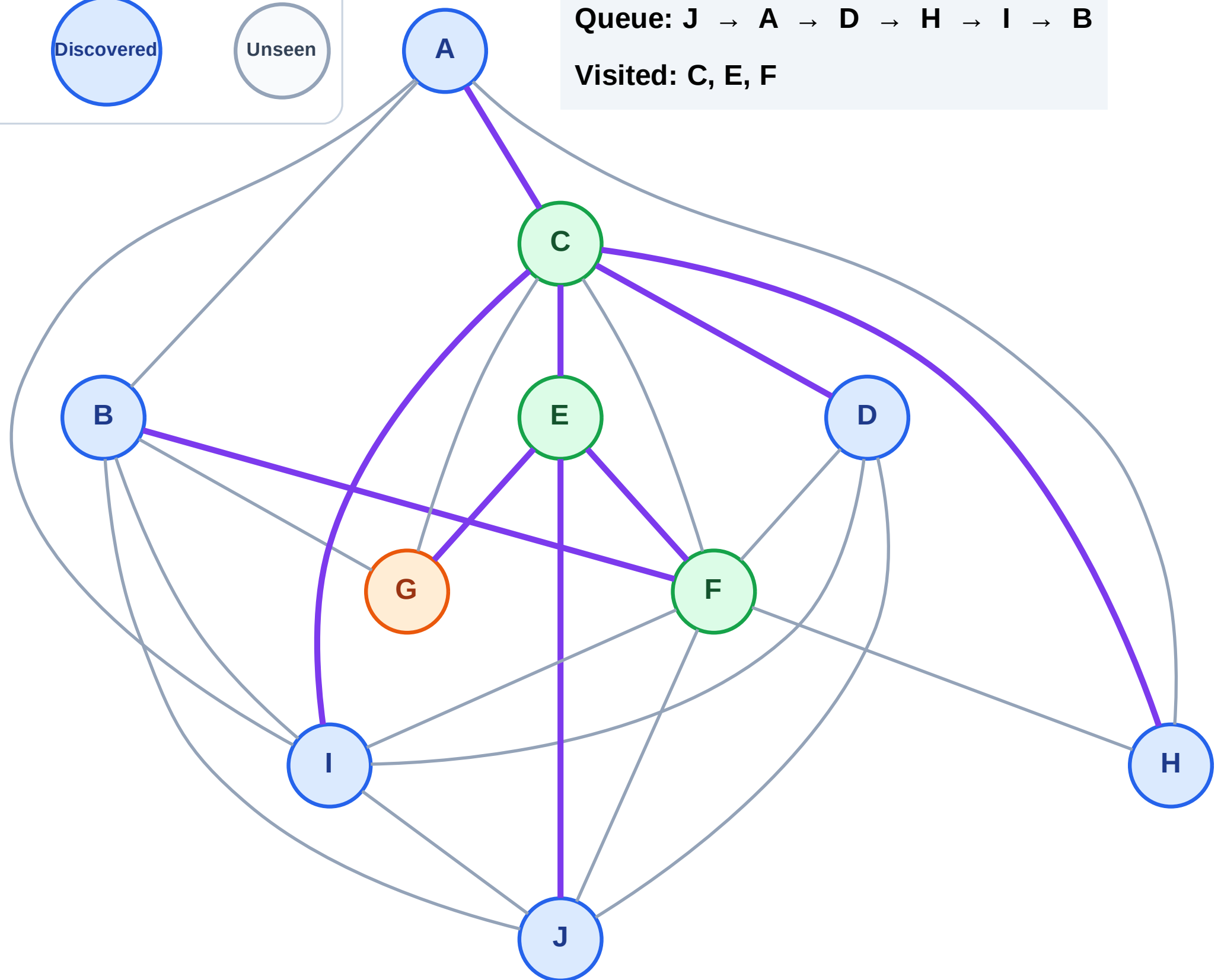
Step 8: Process node G

Legend



Queue: J → A → D → H → I → B

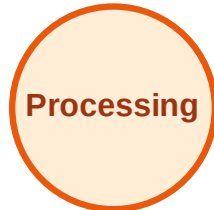
Visited: C, E, F



BFS Traversal

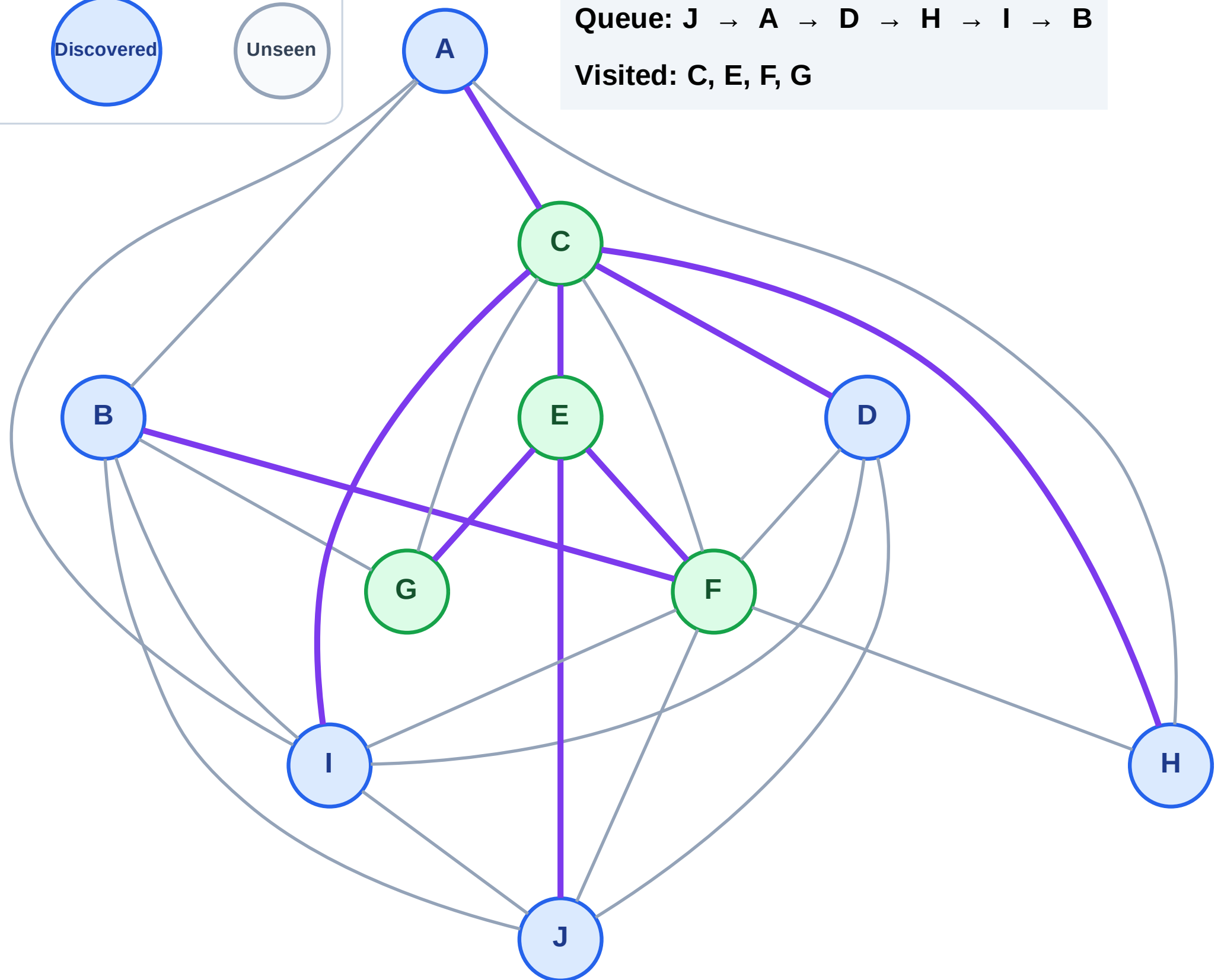
Step 9: No new neighbors from G

Legend



Queue: J → A → D → H → I → B

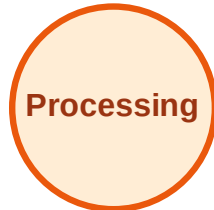
Visited: C, E, F, G



BFS Traversal

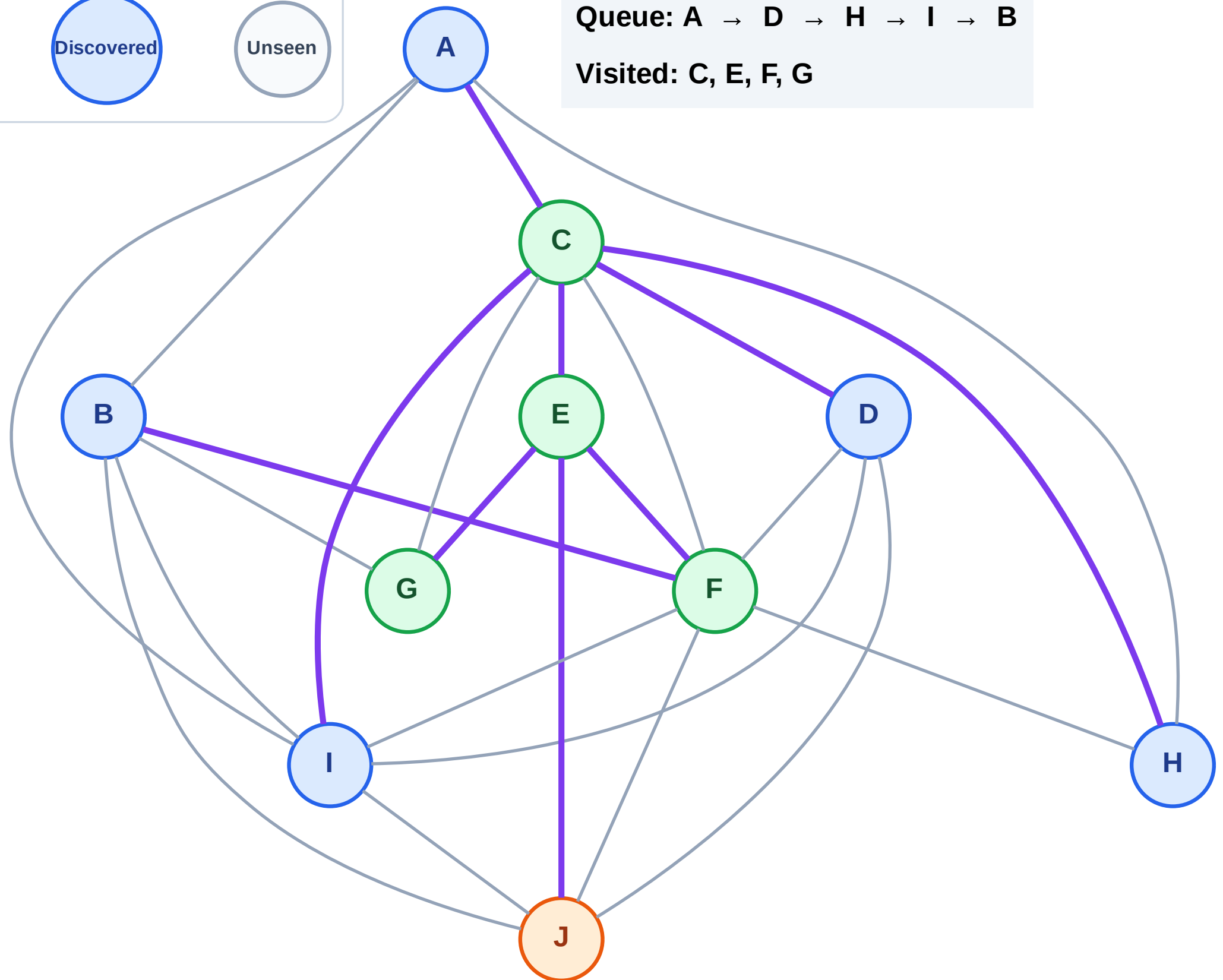
Step 10: Process node J

Legend



Queue: A → D → H → I → B

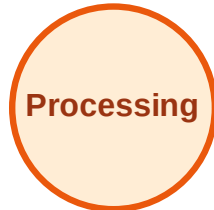
Visited: C, E, F, G



BFS Traversal

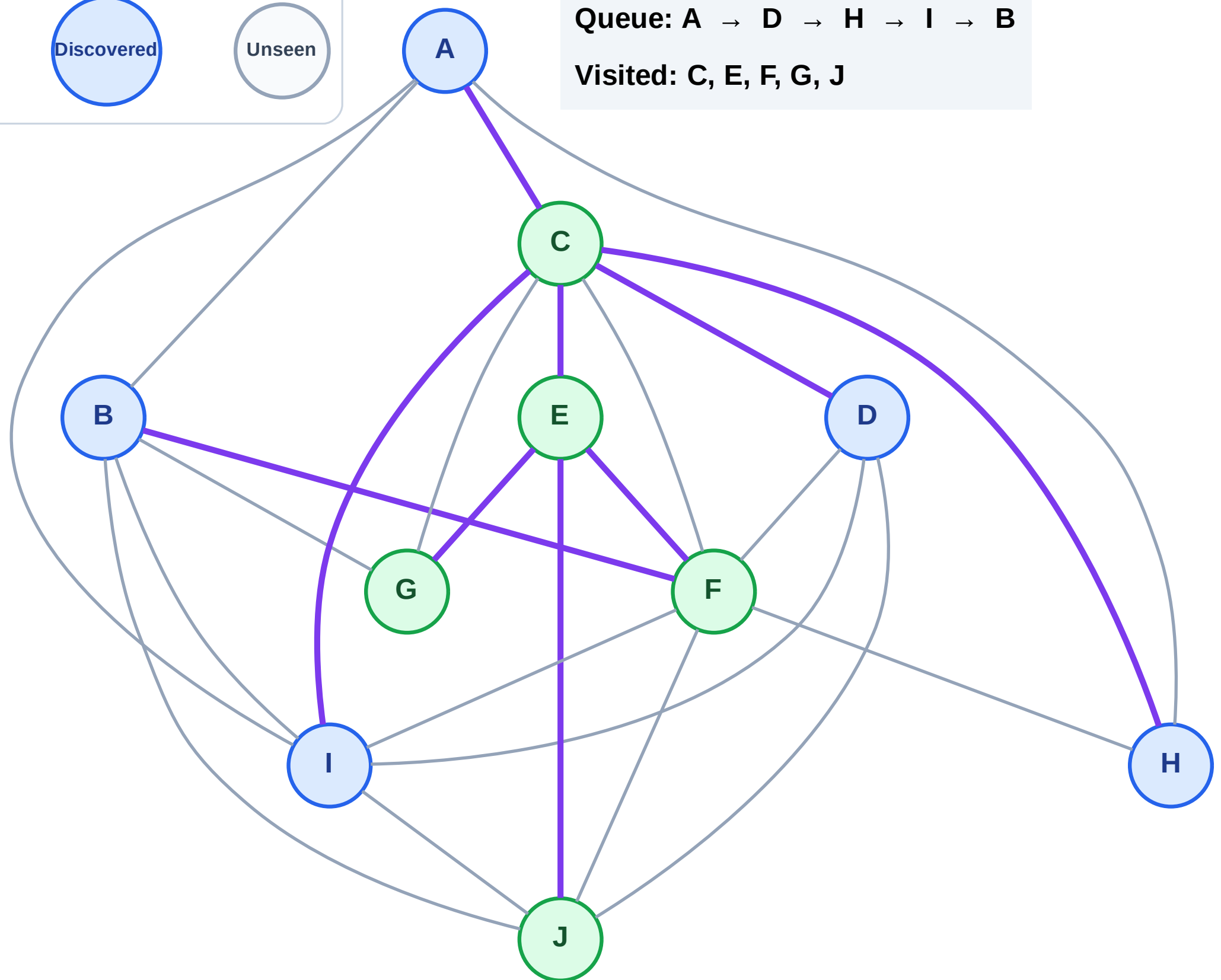
Step 11: No new neighbors from J

Legend



Queue: A → D → H → I → B

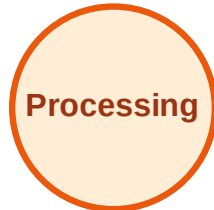
Visited: C, E, F, G, J



BFS Traversal

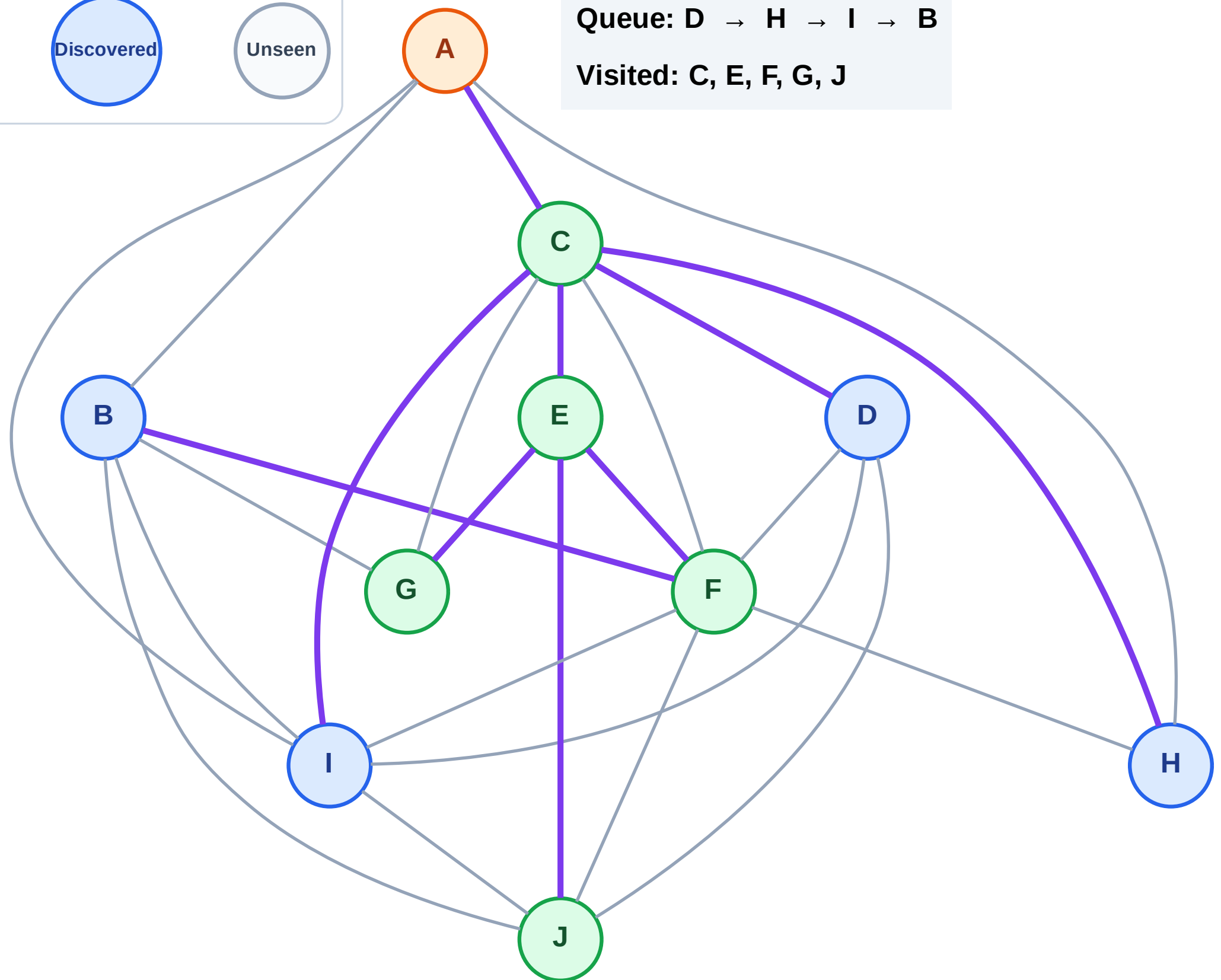
Step 12: Process node A

Legend



Queue: D → H → I → B

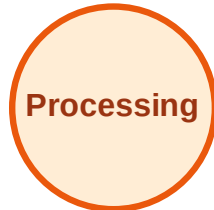
Visited: C, E, F, G, J



BFS Traversal

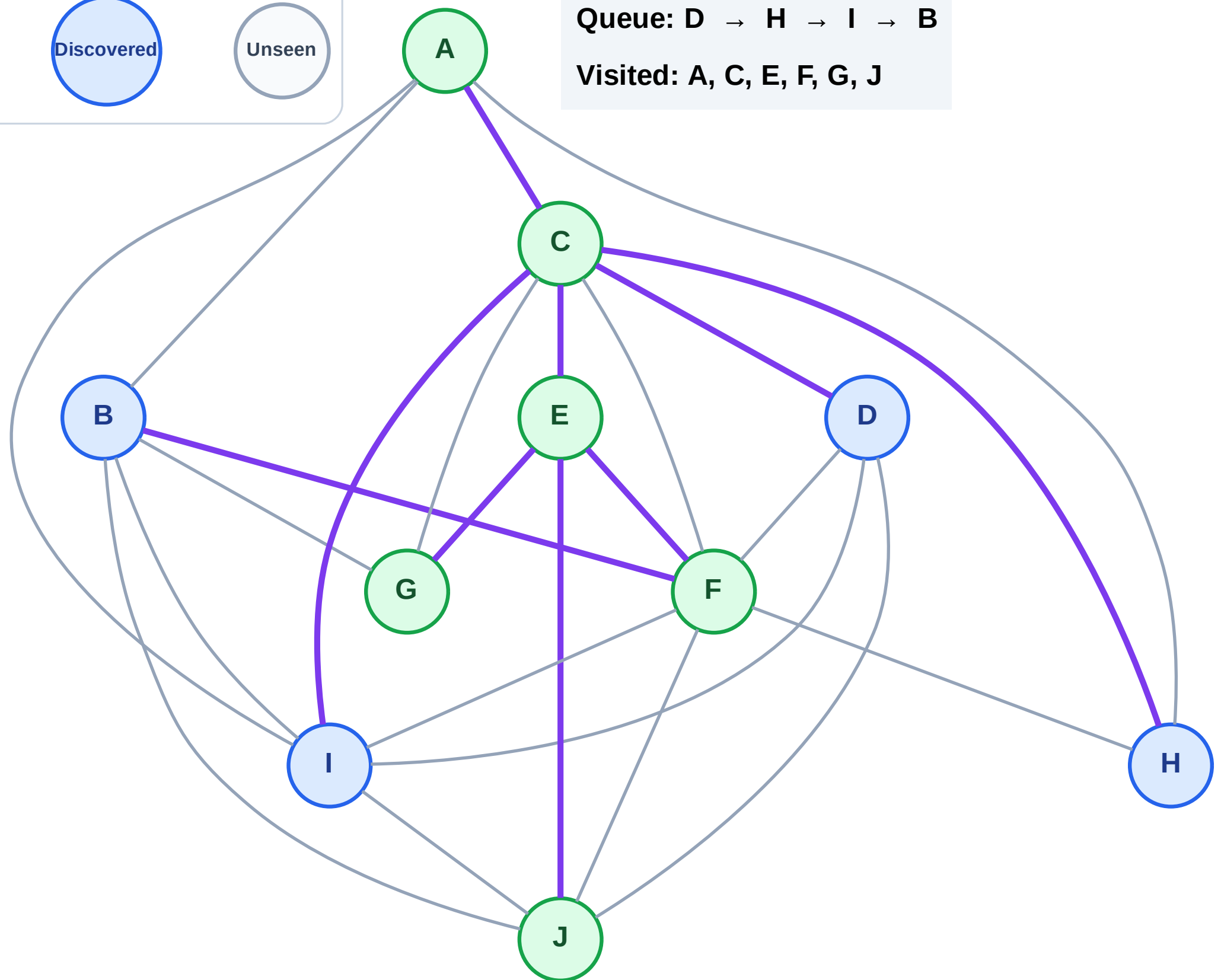
Step 13: No new neighbors from A

Legend



Queue: D → H → I → B

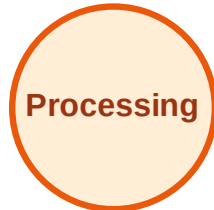
Visited: A, C, E, F, G, J



BFS Traversal

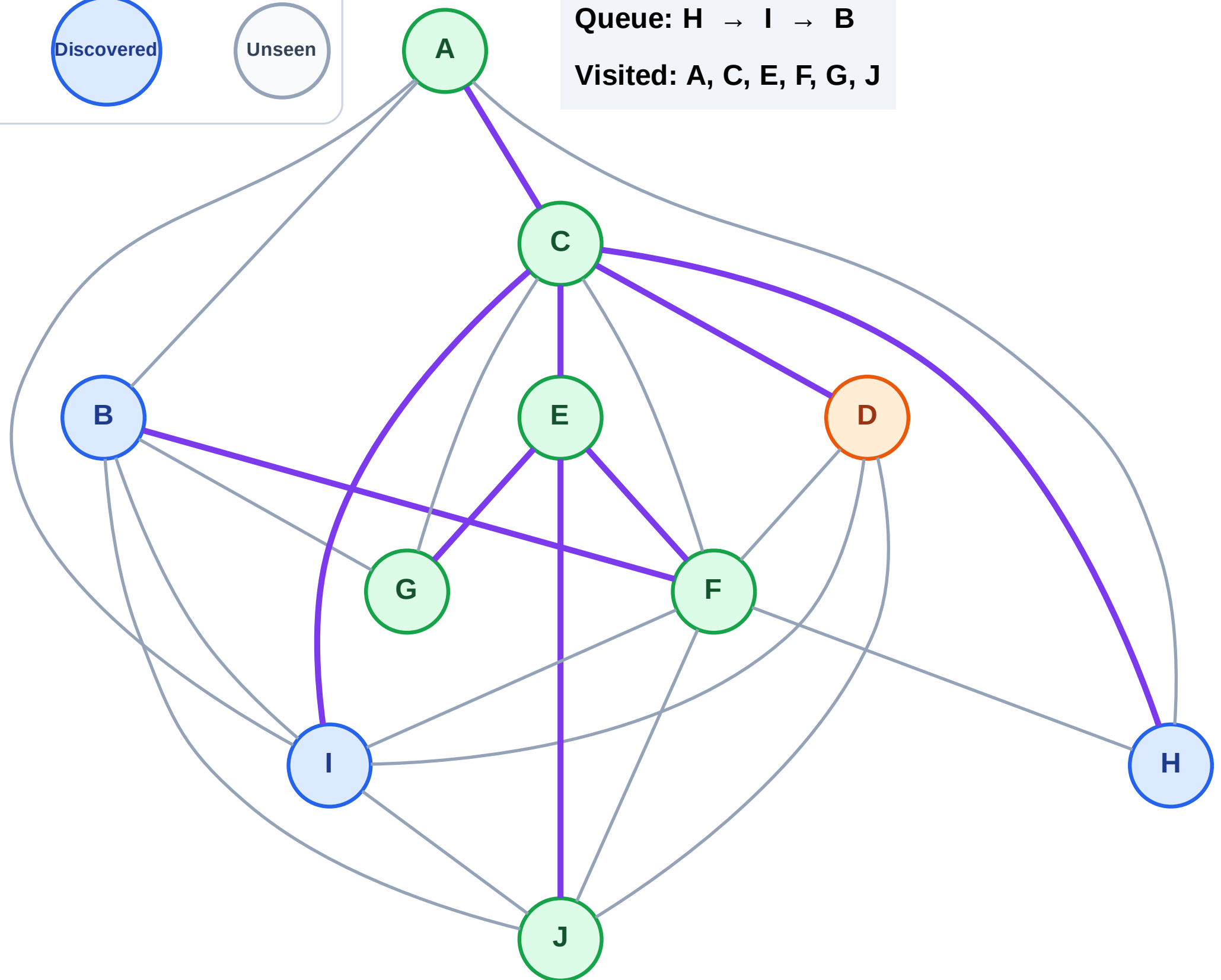
Step 14: Process node D

Legend



Queue: H → I → B

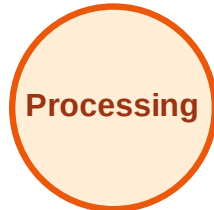
Visited: A, C, E, F, G, J



BFS Traversal

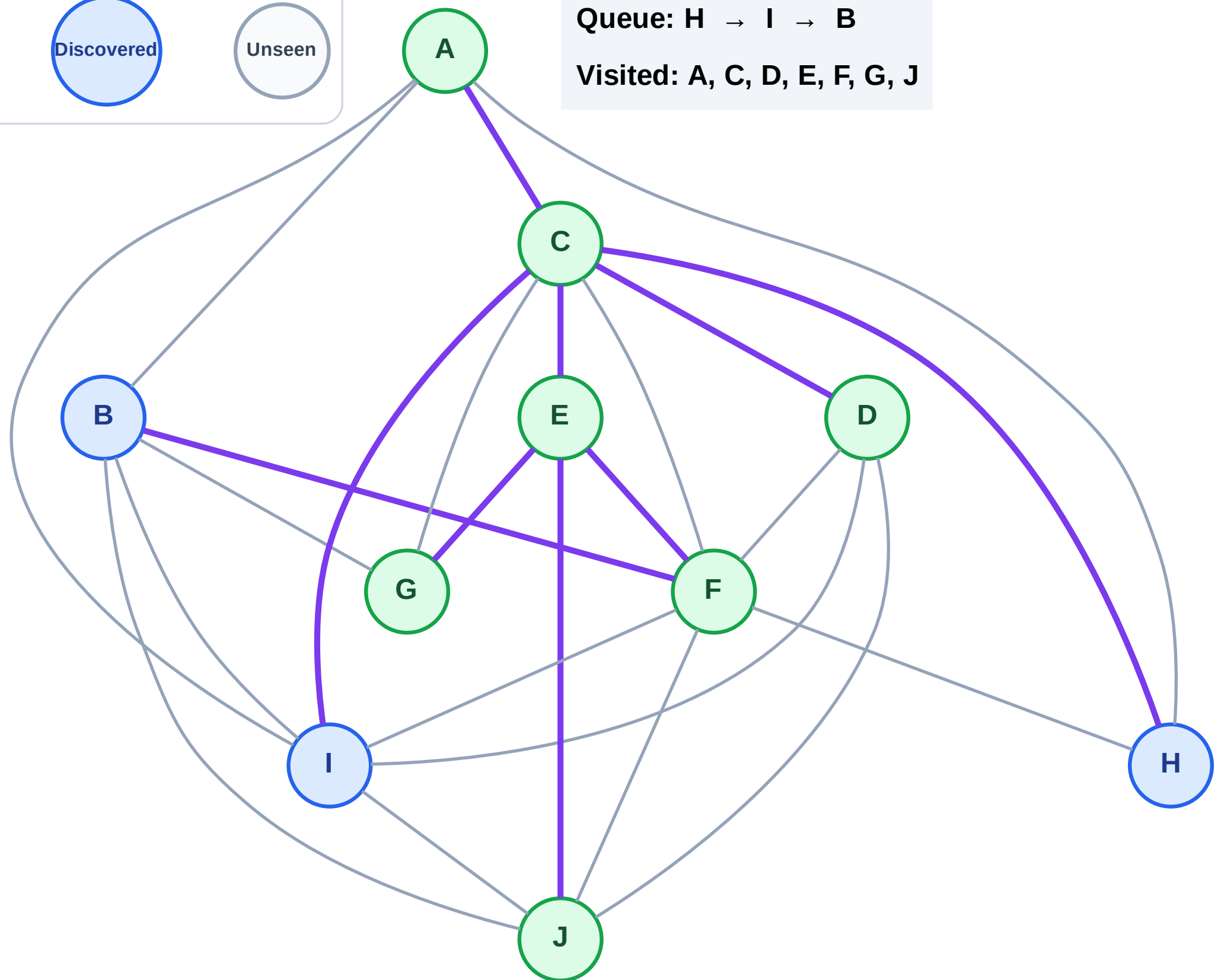
Step 15: No new neighbors from D

Legend



Queue: H → I → B

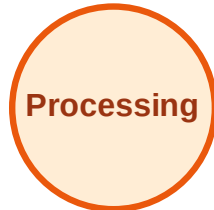
Visited: A, C, D, E, F, G, J



BFS Traversal

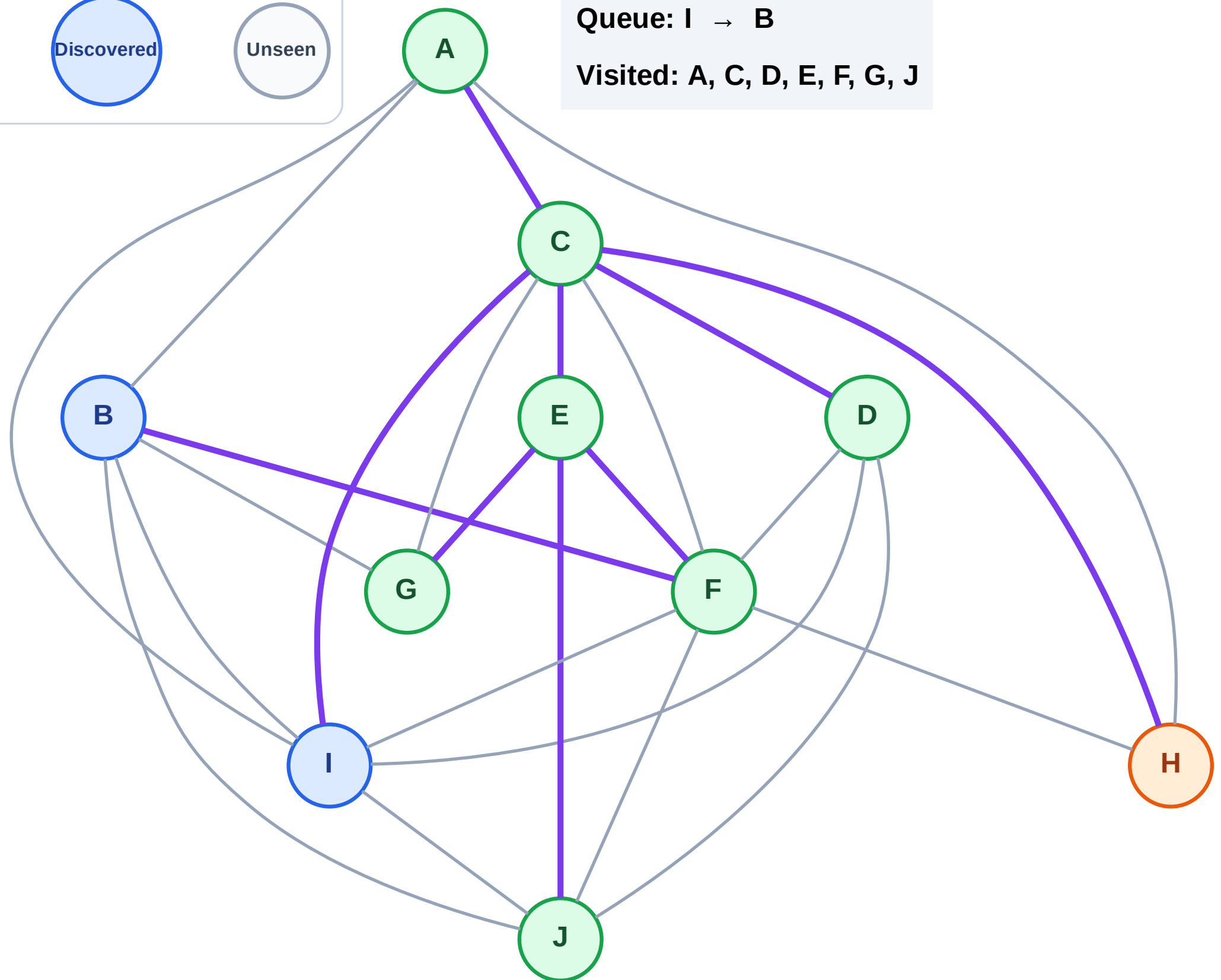
Step 16: Process node H

Legend



Queue: I → B

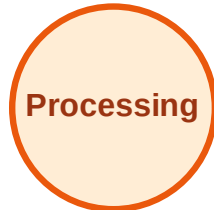
Visited: A, C, D, E, F, G, J



BFS Traversal

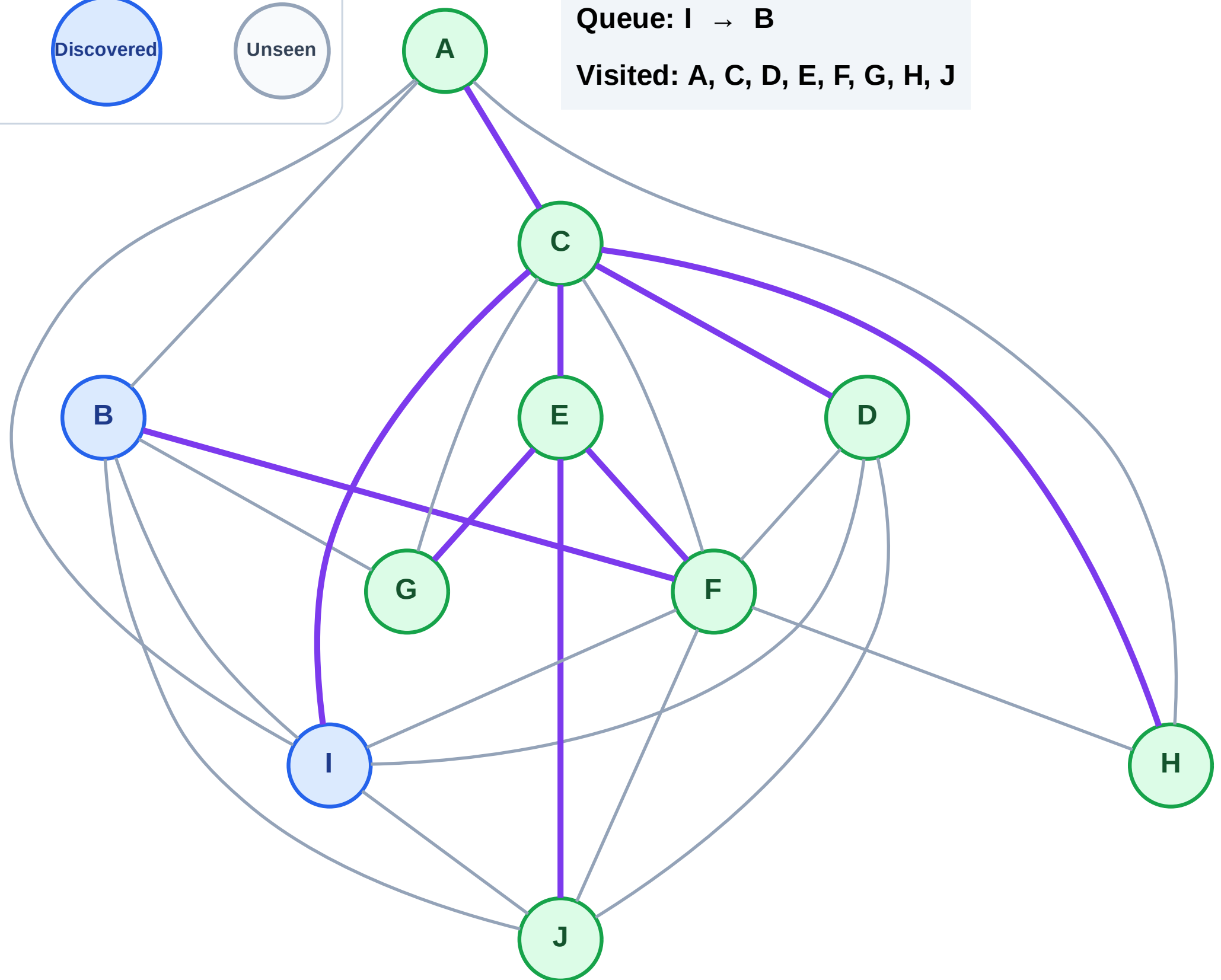
Step 17: No new neighbors from H

Legend



Queue: I → B

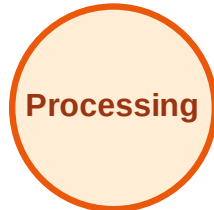
Visited: A, C, D, E, F, G, H, J



BFS Traversal

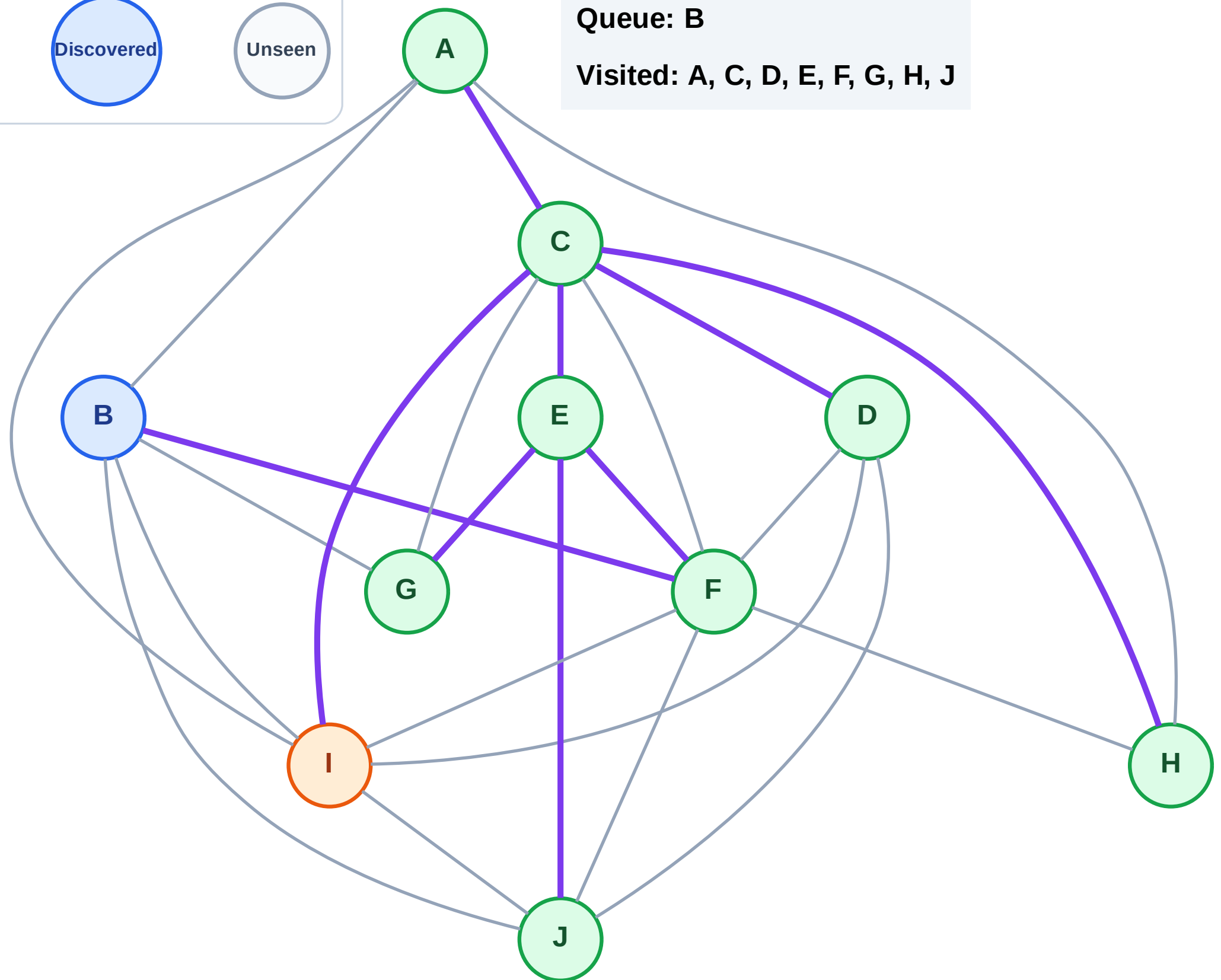
Step 18: Process node I

Legend



Queue: B

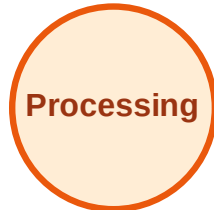
Visited: A, C, D, E, F, G, H, J



BFS Traversal

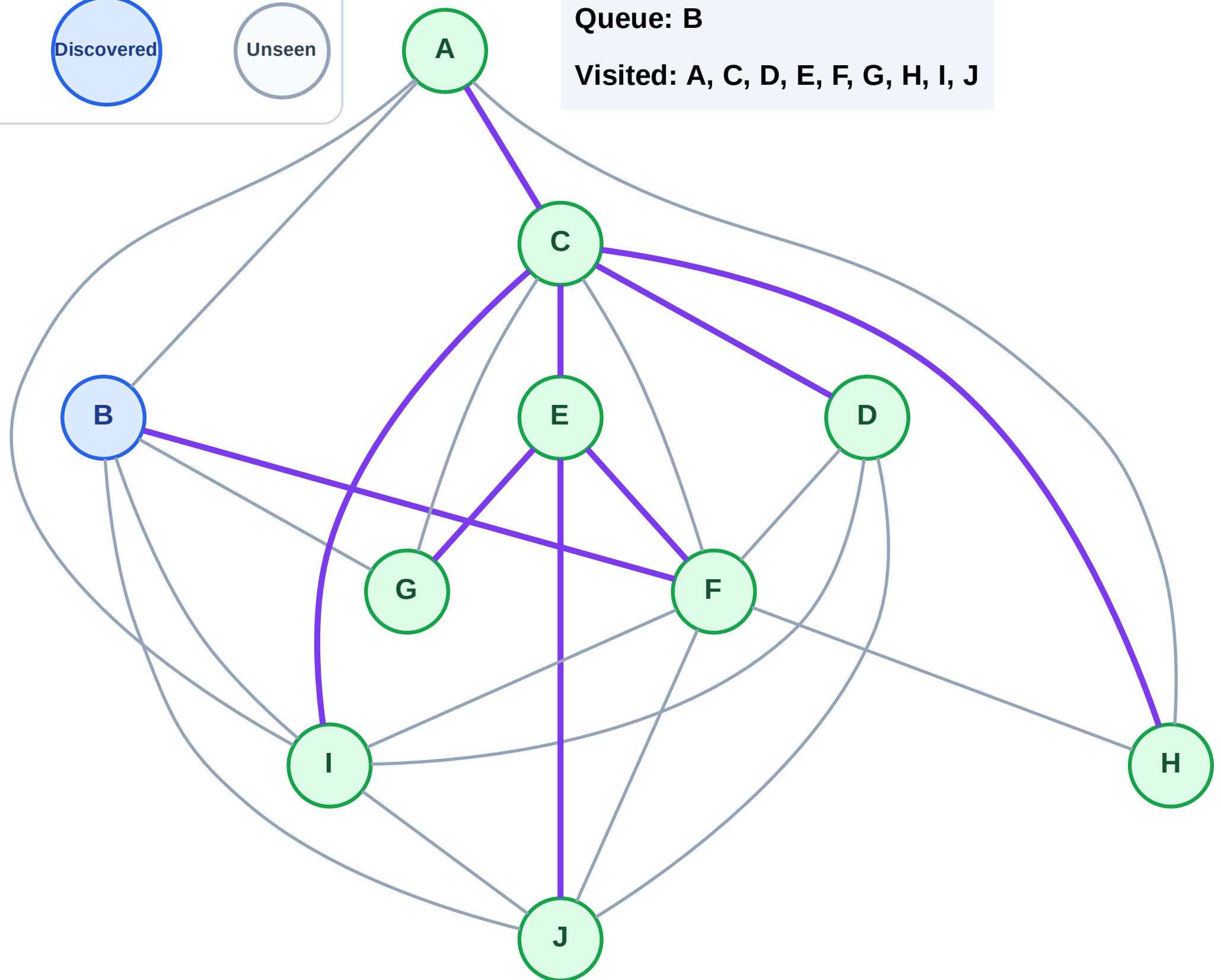
Step 19: No new neighbors from I

Legend



Queue: B

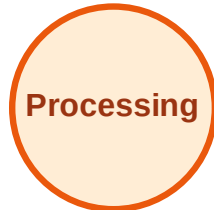
Visited: A, C, D, E, F, G, H, I, J



BFS Traversal

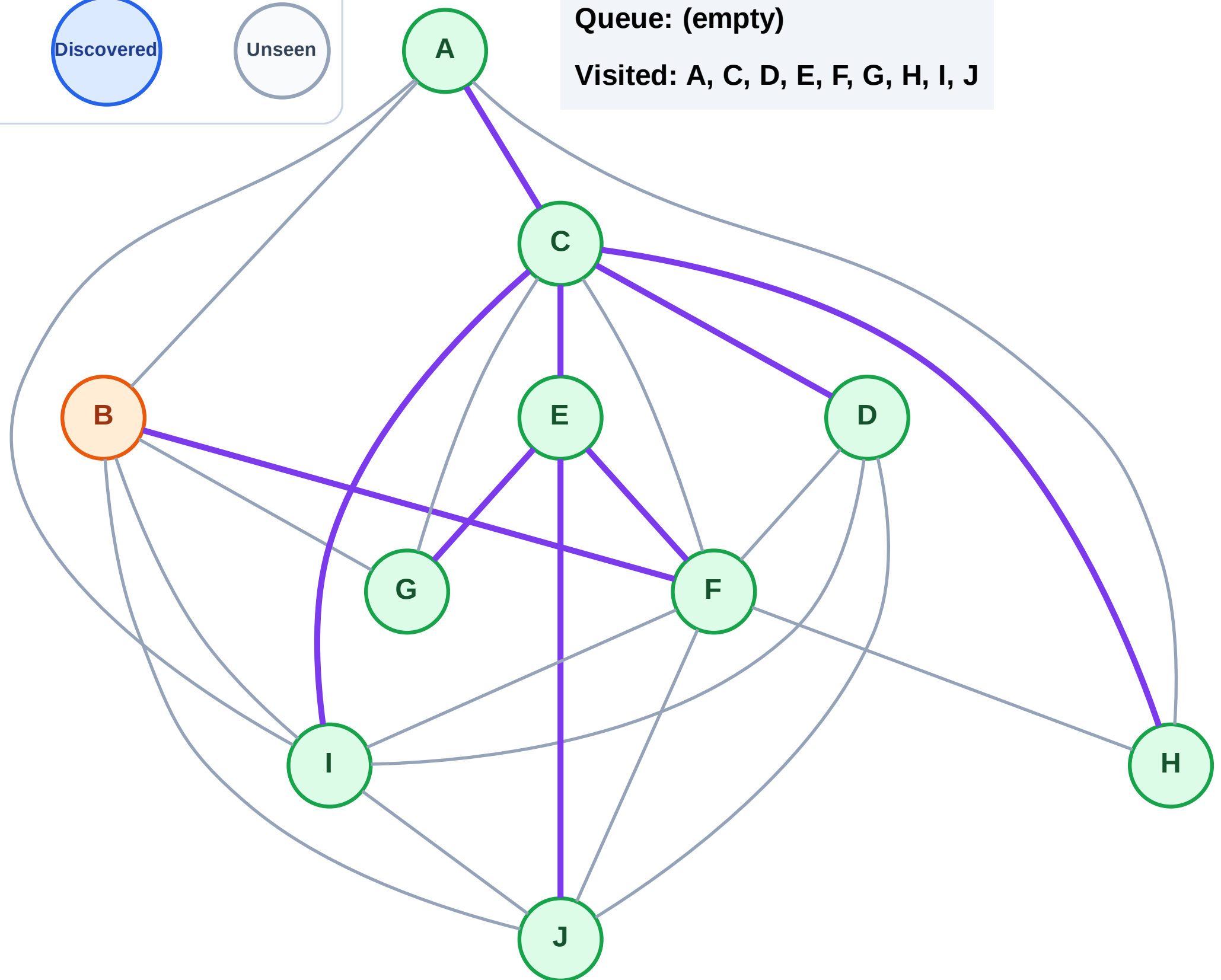
Step 20: Process node B

Legend



Queue: (empty)

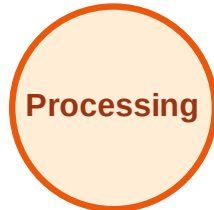
Visited: A, C, D, E, F, G, H, I, J



BFS Traversal

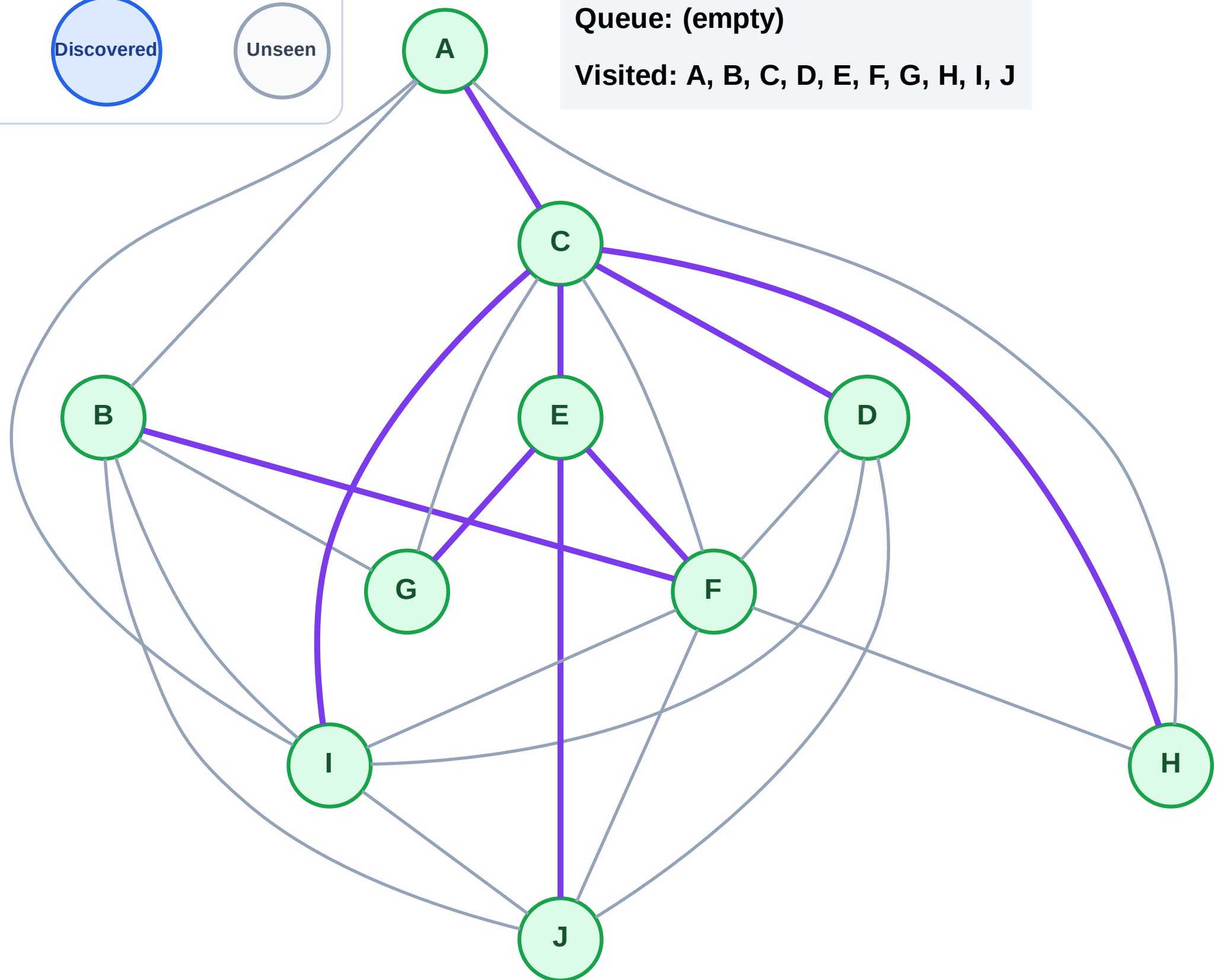
Step 21: No new neighbors from B

Legend



Queue: (empty)

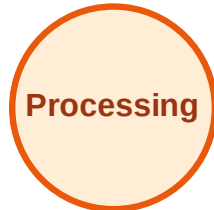
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

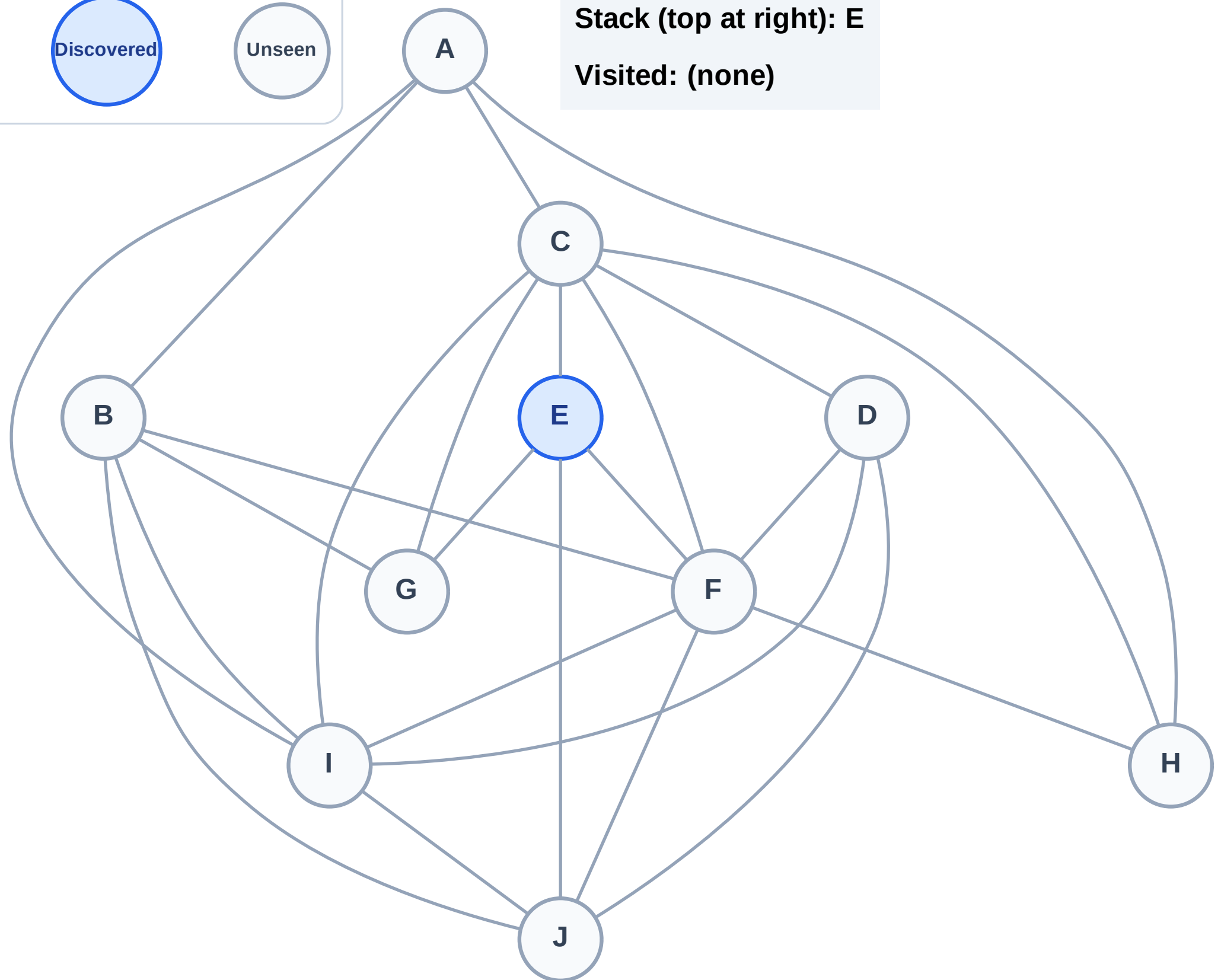
Step 1: Start at E

Legend



Stack (top at right): E

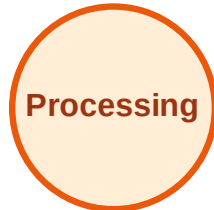
Visited: (none)



DFS Traversal

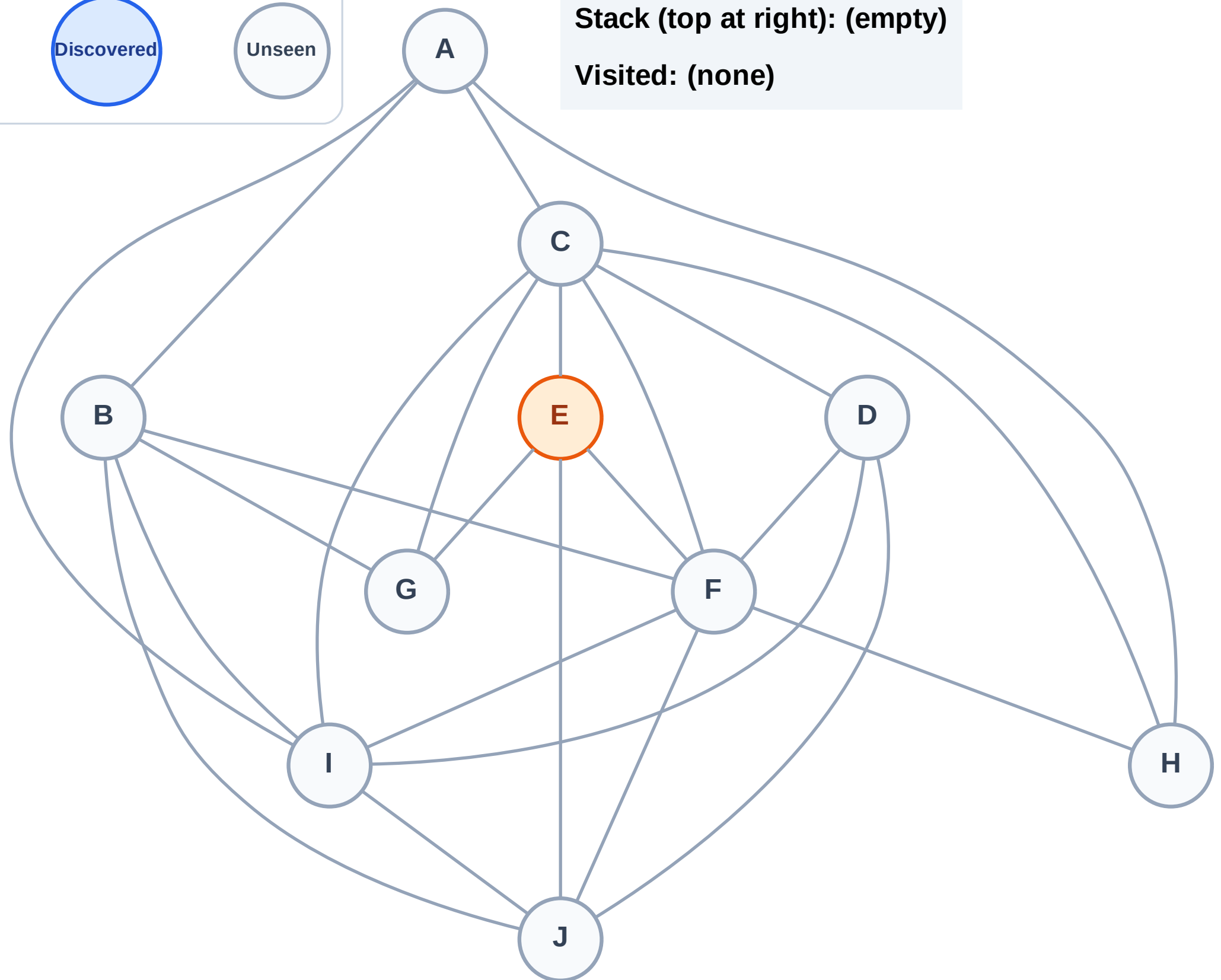
Step 2: Process node E

Legend



Stack (top at right): (empty)

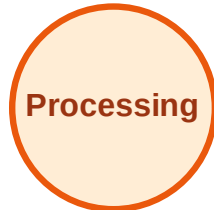
Visited: (none)



DFS Traversal

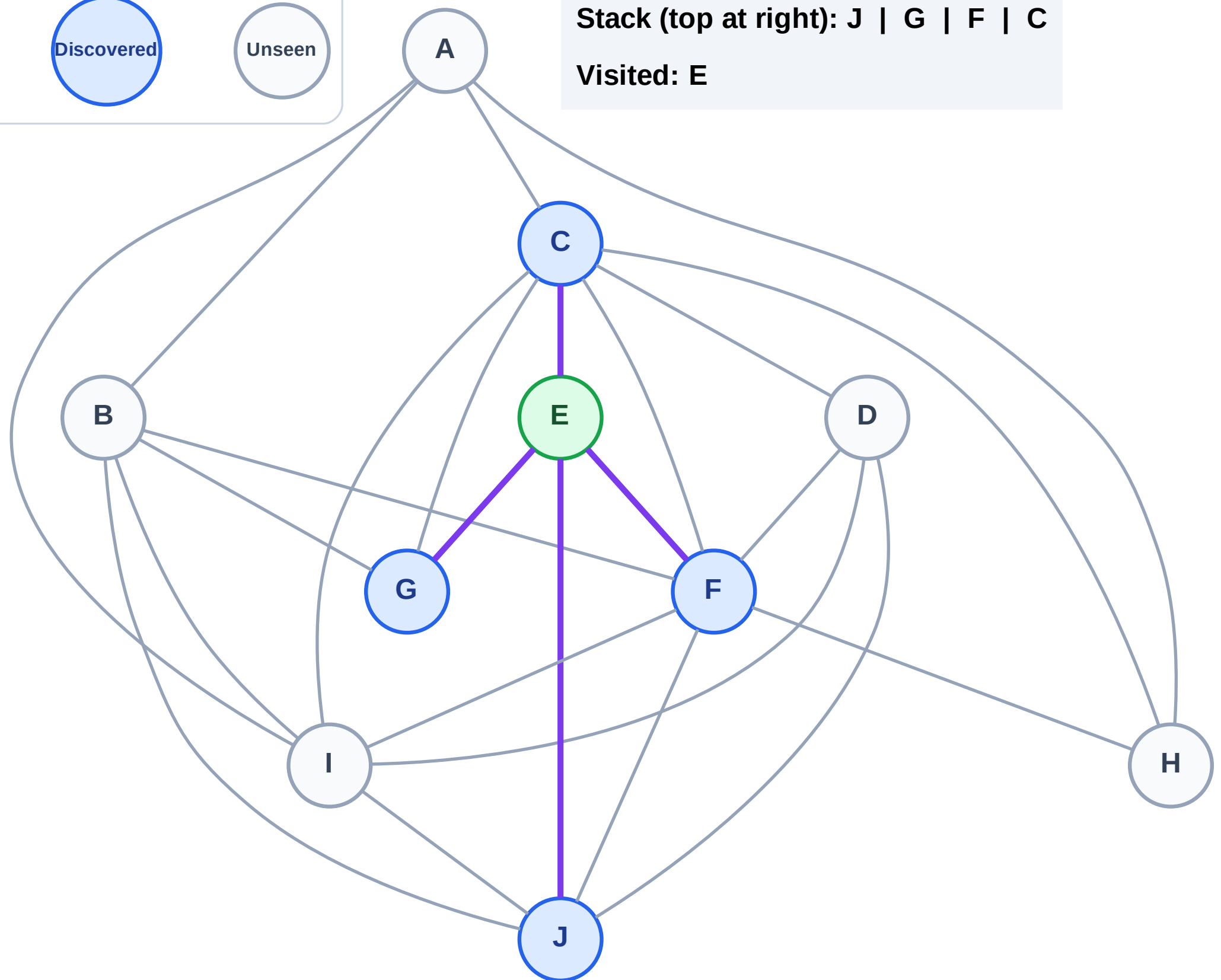
Step 3: Discover J, G, F, C from E

Legend



Stack (top at right): J | G | F | C

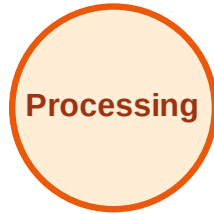
Visited: E



DFS Traversal

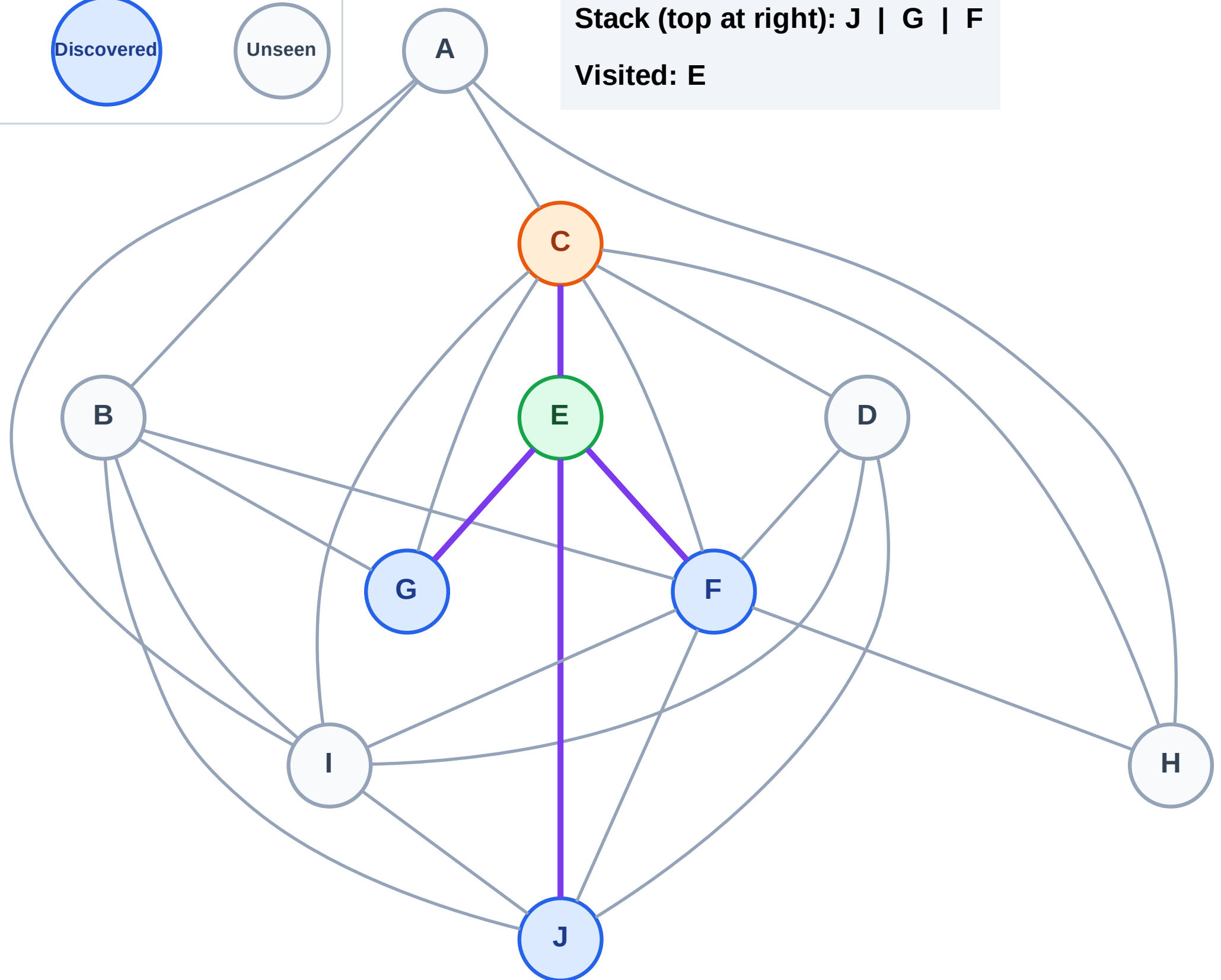
Step 4: Process node C

Legend



Stack (top at right): J | G | F

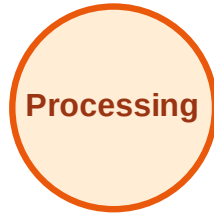
Visited: E



DFS Traversal

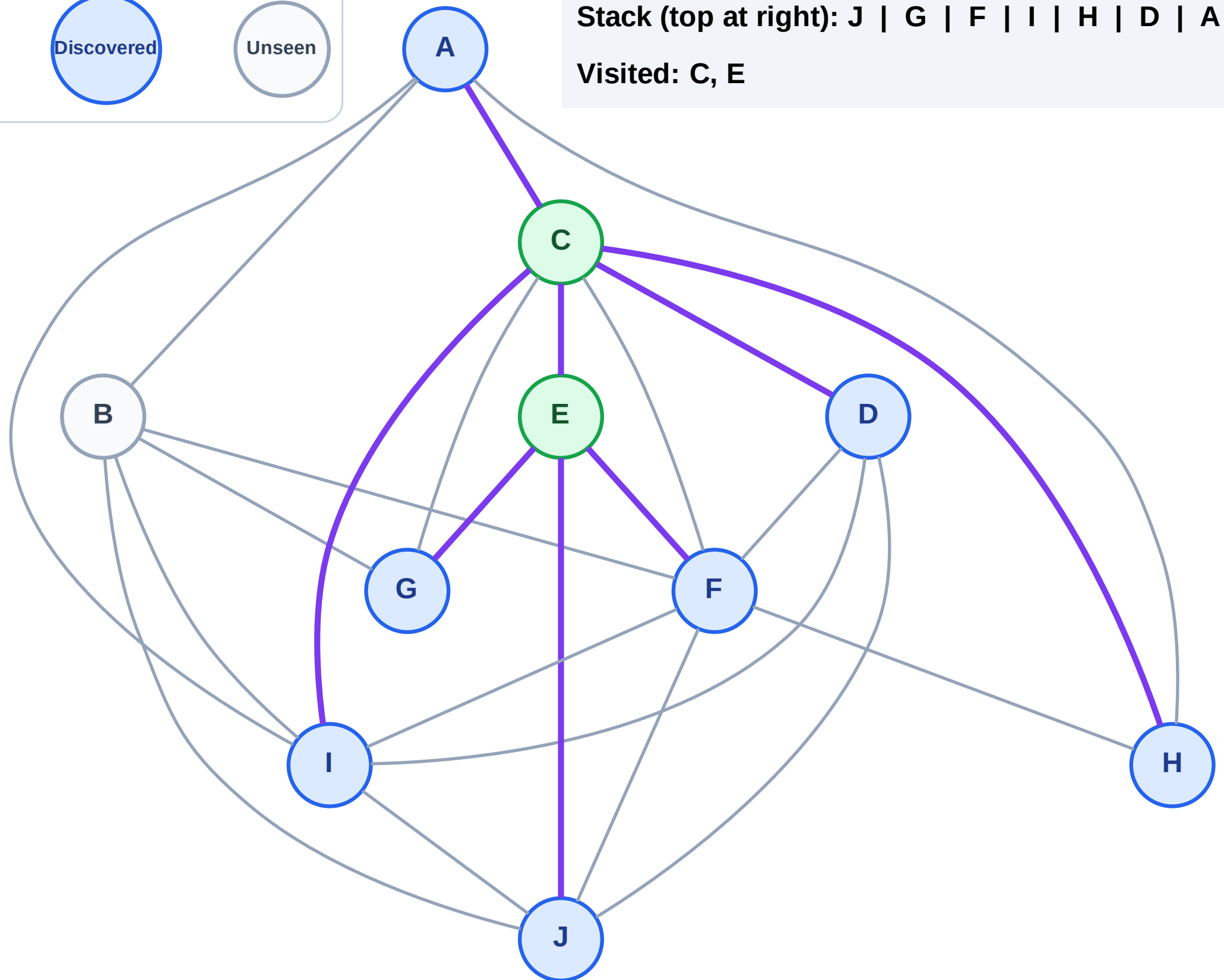
Step 5: Discover I, H, D, A from C

Legend



Stack (top at right): J | G | F | I | H | D | A

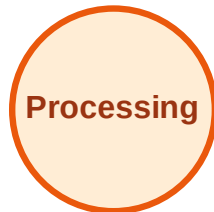
Visited: C, E



DFS Traversal

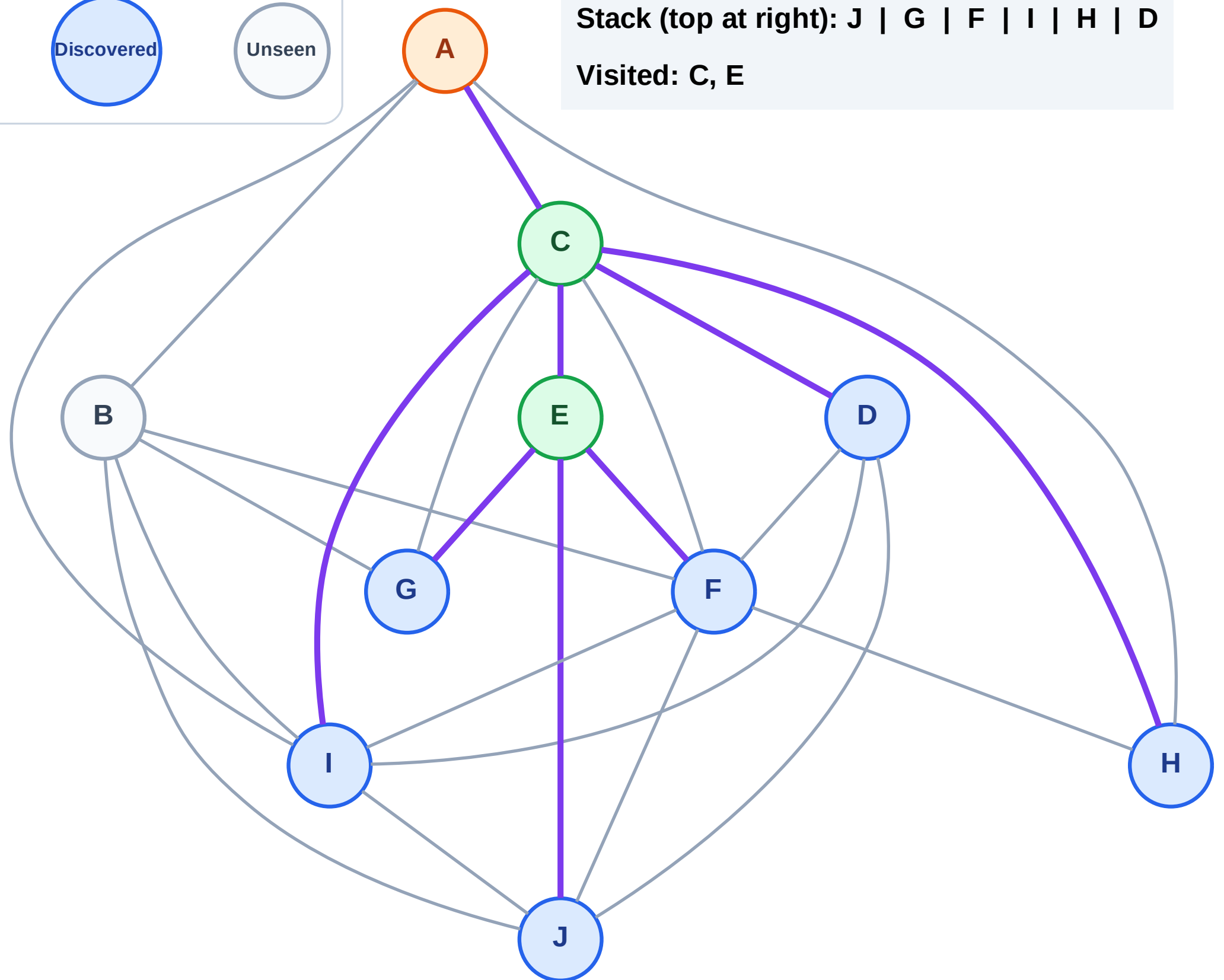
Step 6: Process node A

Legend



Stack (top at right): J | G | F | I | H | D

Visited: C, E



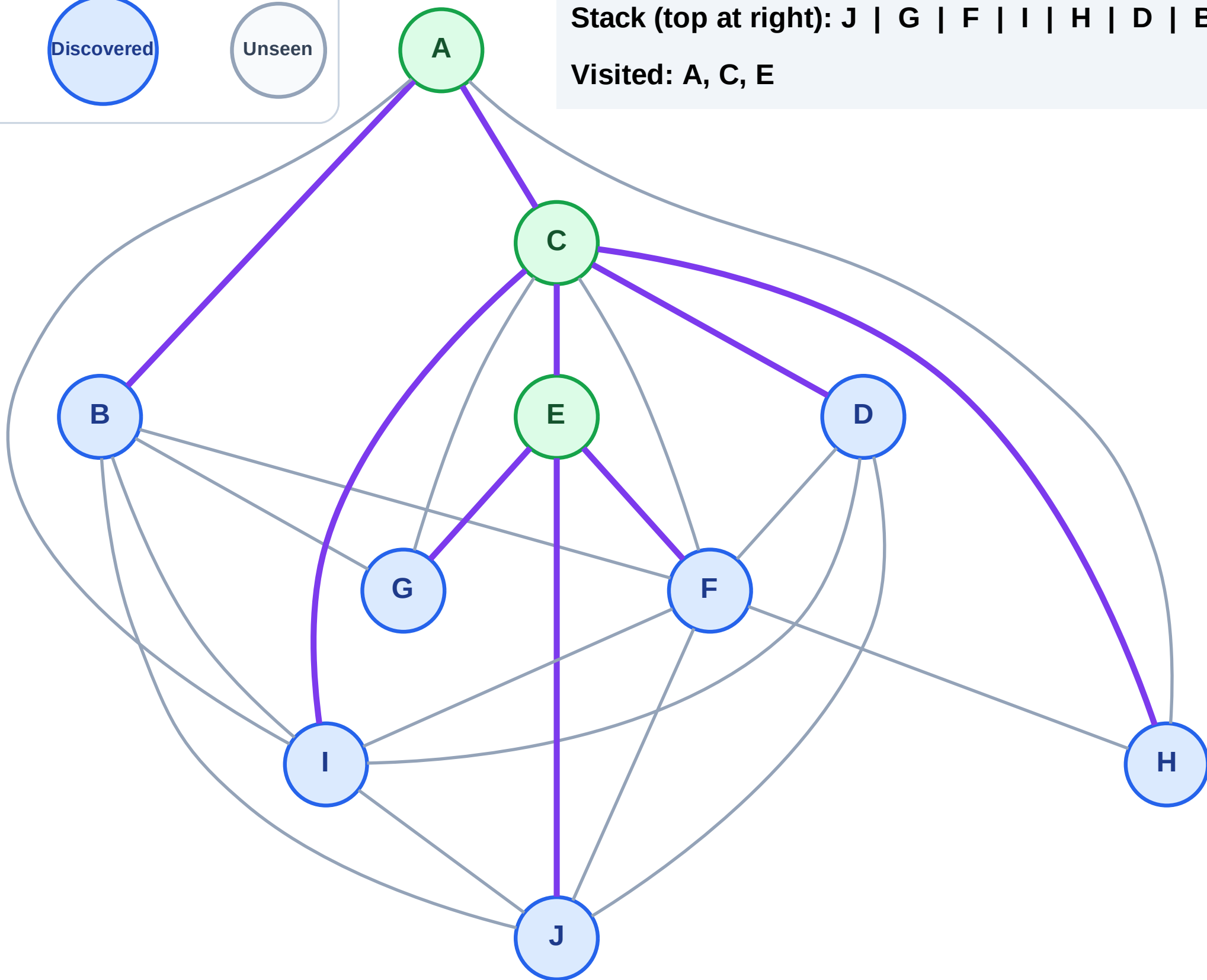
Step 7: Discover B from A

Step 7: Discover B from A



Stack (top at right): J | G | F | I | H | D | B

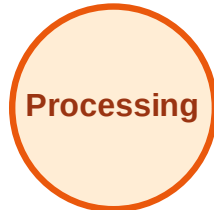
Visited: A, C, E



DFS Traversal

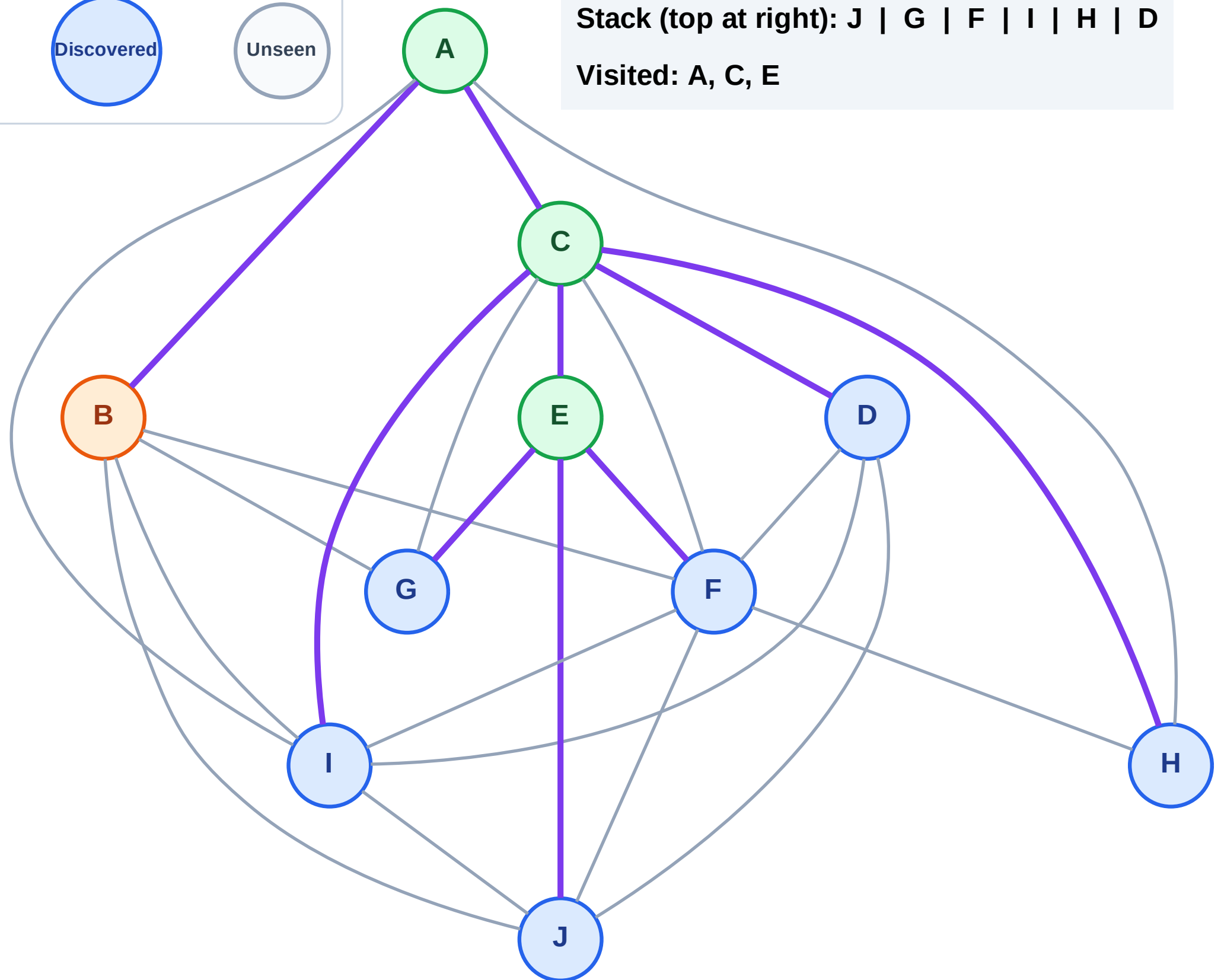
Step 8: Process node B

Legend



Stack (top at right): J | G | F | I | H | D

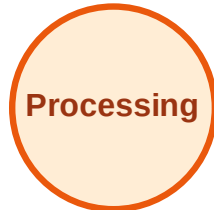
Visited: A, C, E



DFS Traversal

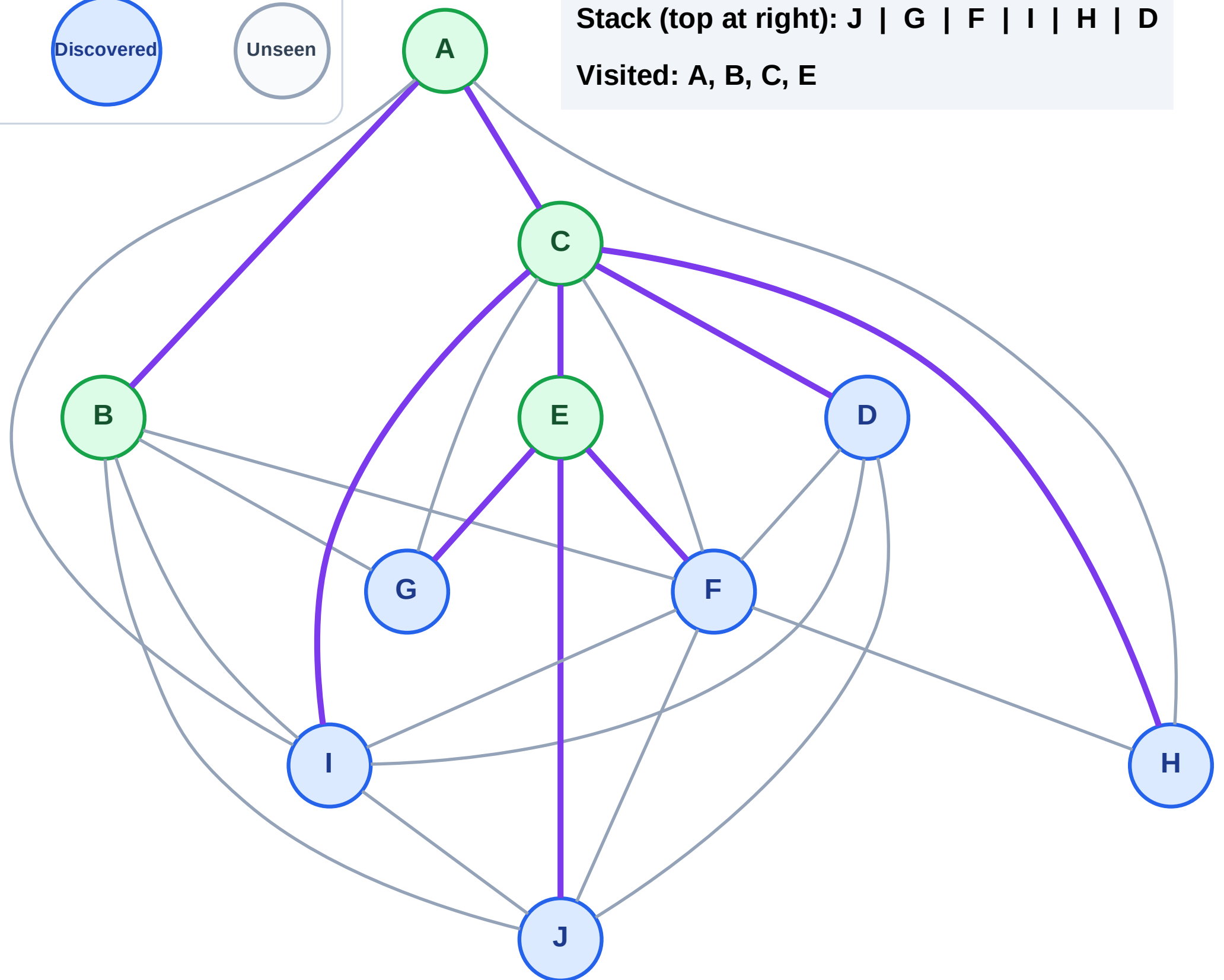
Step 9: No new neighbors from B

Legend



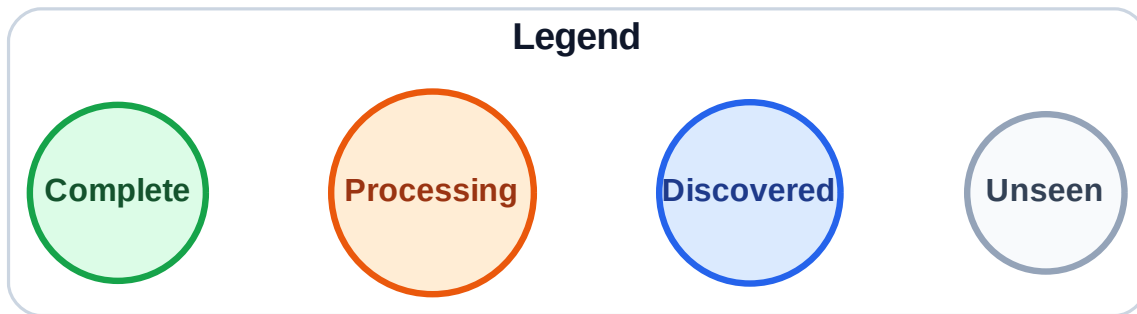
Stack (top at right): J | G | F | I | H | D

Visited: A, B, C, E



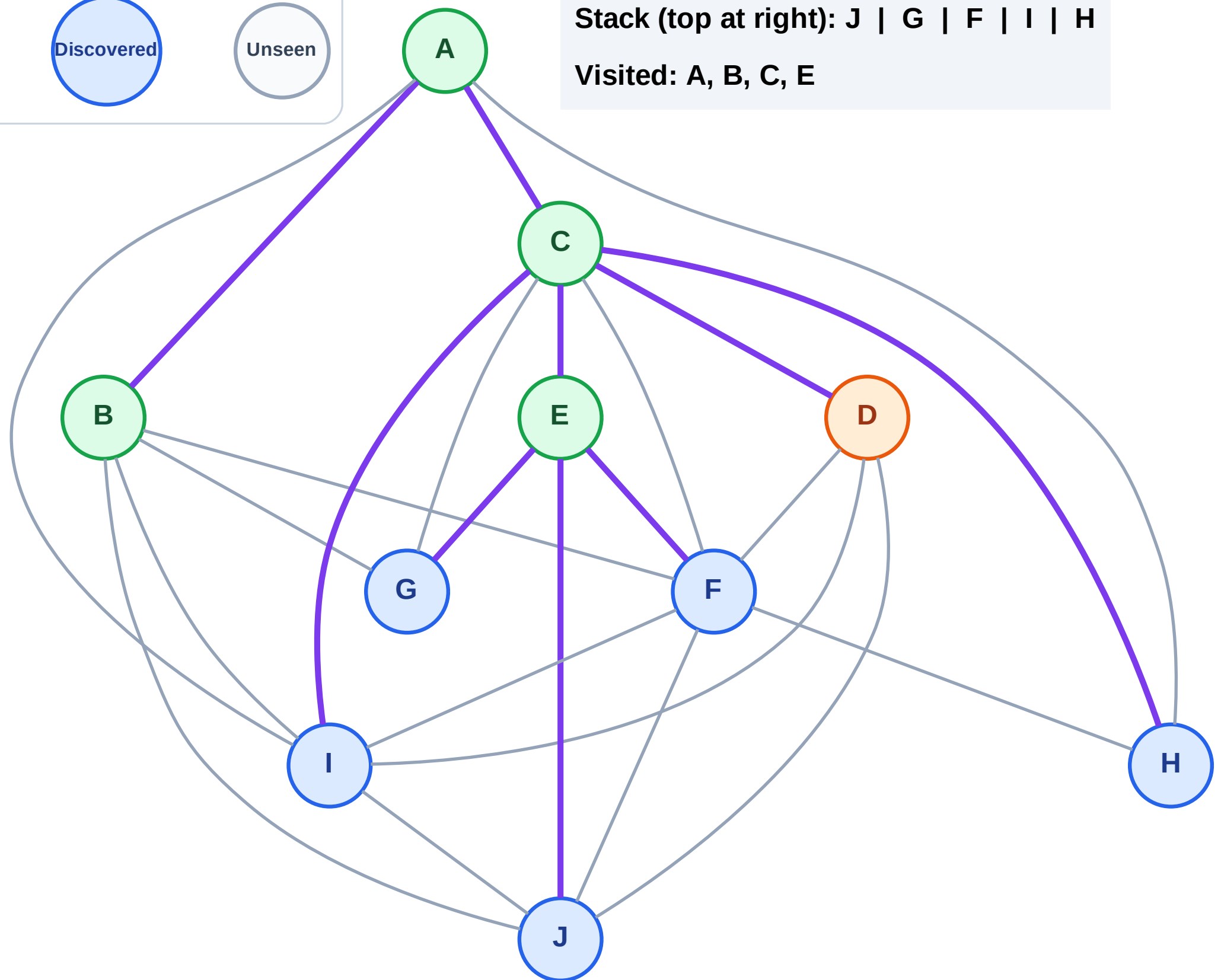
Step 10: Process node D

Step 10: Process node D



Stack (top at right): J | G | F | I | H

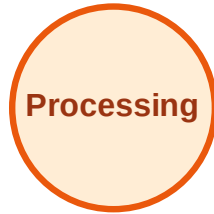
Visited: A, B, C, E



DFS Traversal

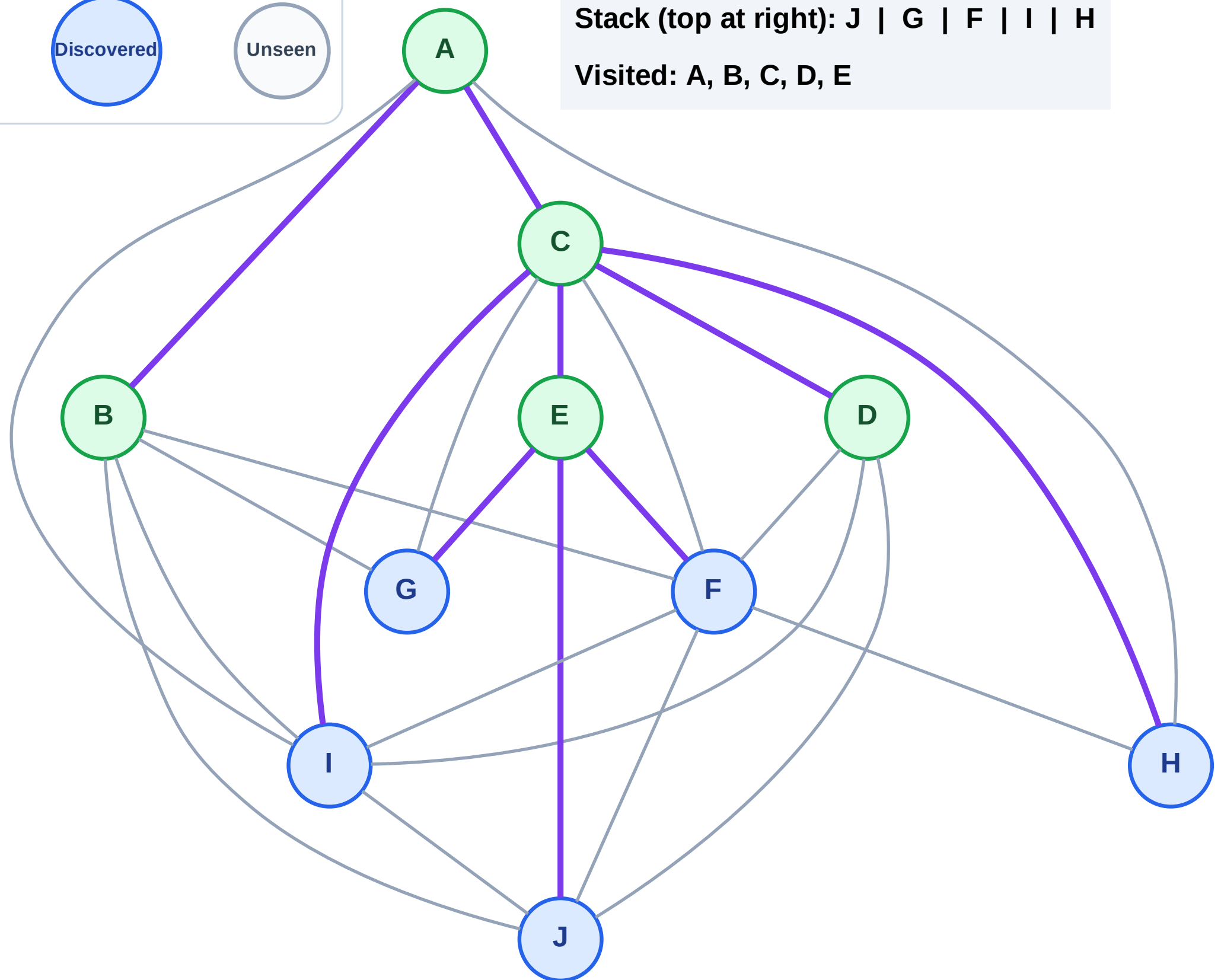
Step 11: No new neighbors from D

Legend



Stack (top at right): J | G | F | I | H

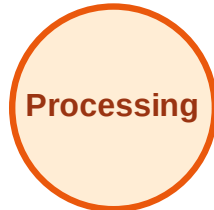
Visited: A, B, C, D, E



DFS Traversal

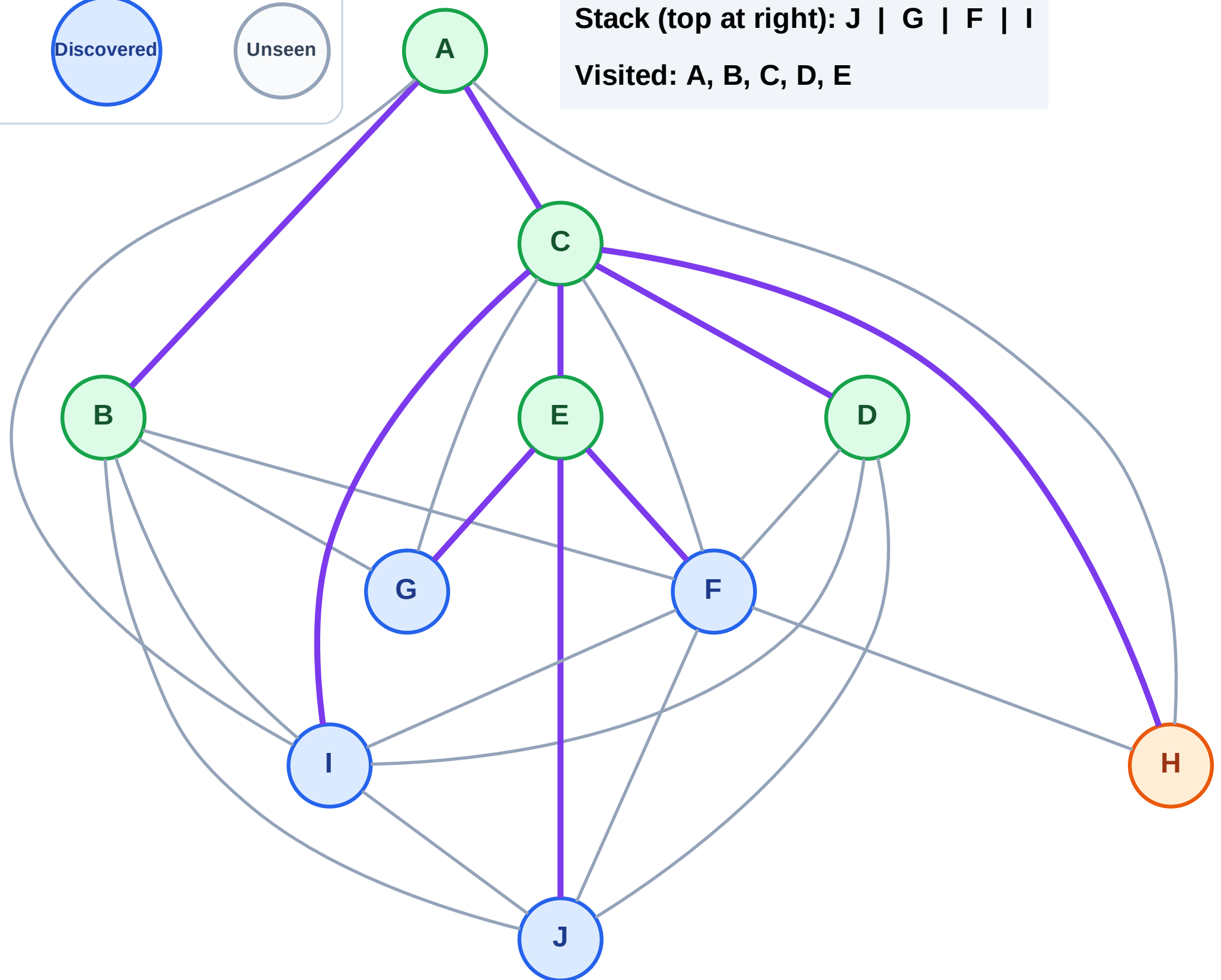
Step 12: Process node H

Legend



Stack (top at right): J | G | F | I

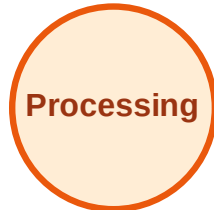
Visited: A, B, C, D, E



DFS Traversal

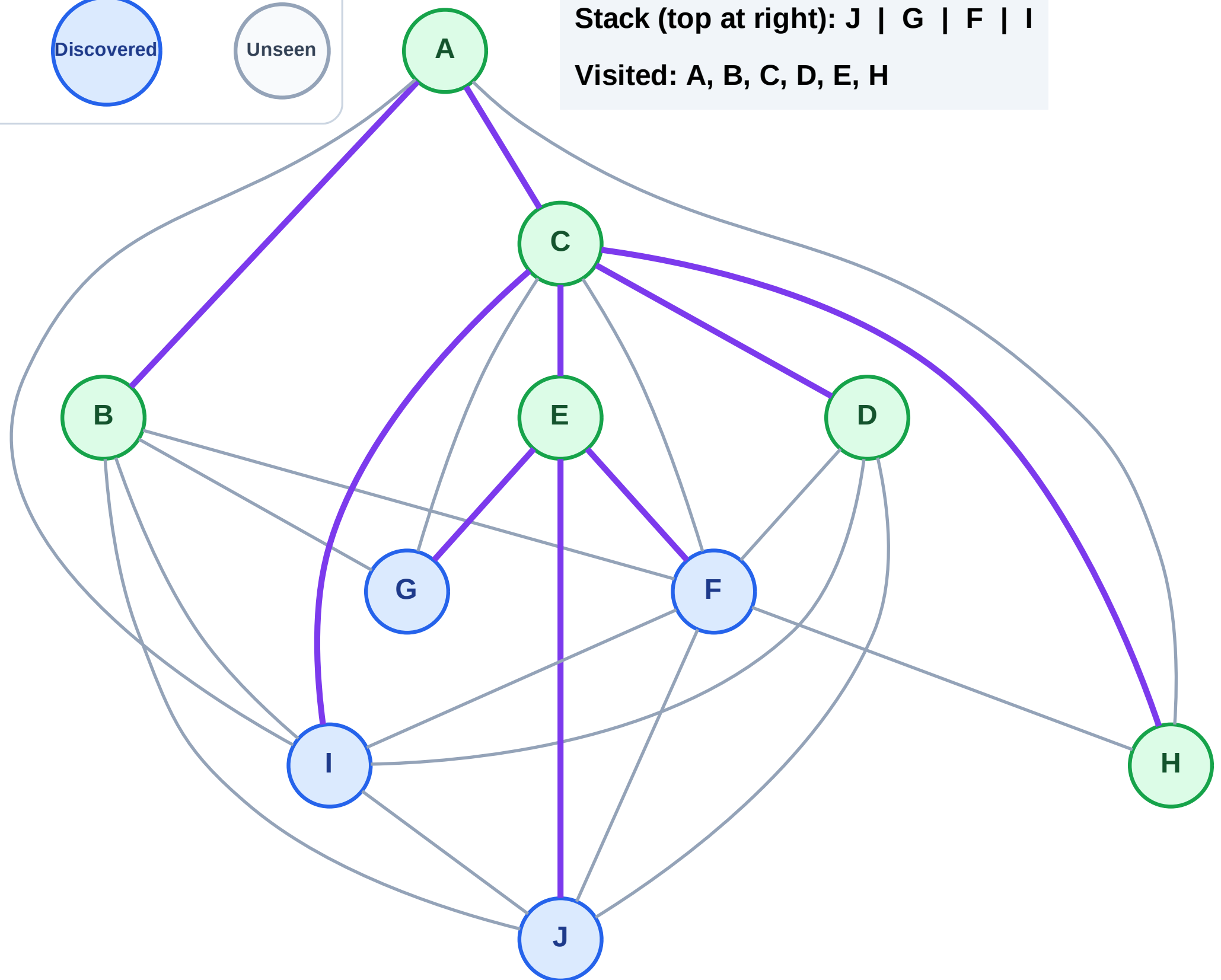
Step 13: No new neighbors from H

Legend



Stack (top at right): J | G | F | I

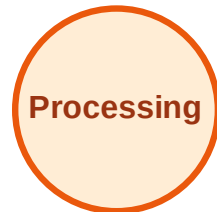
Visited: A, B, C, D, E, H



DFS Traversal

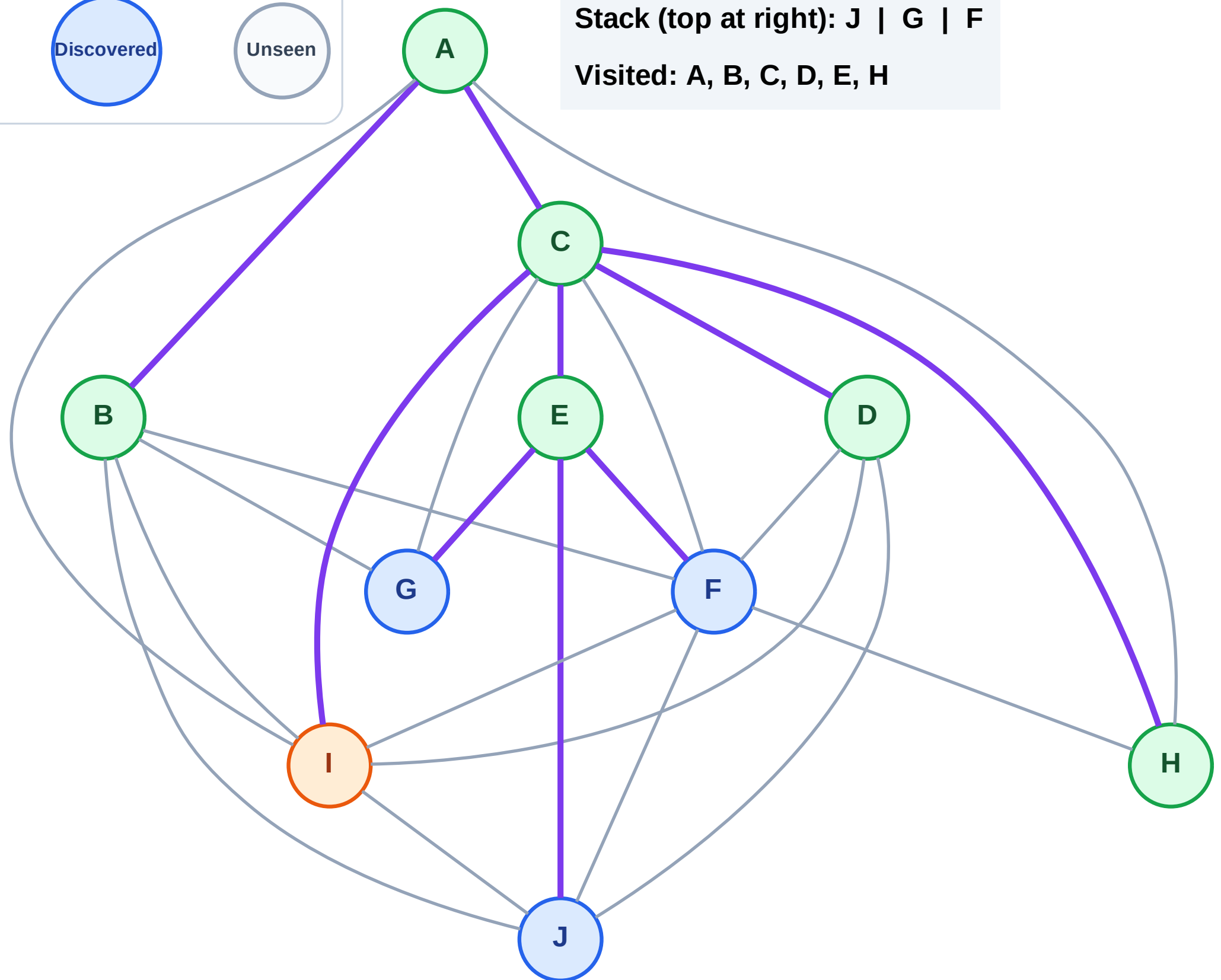
Step 14: Process node I

Legend



Stack (top at right): J | G | F

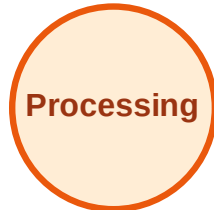
Visited: A, B, C, D, E, H



DFS Traversal

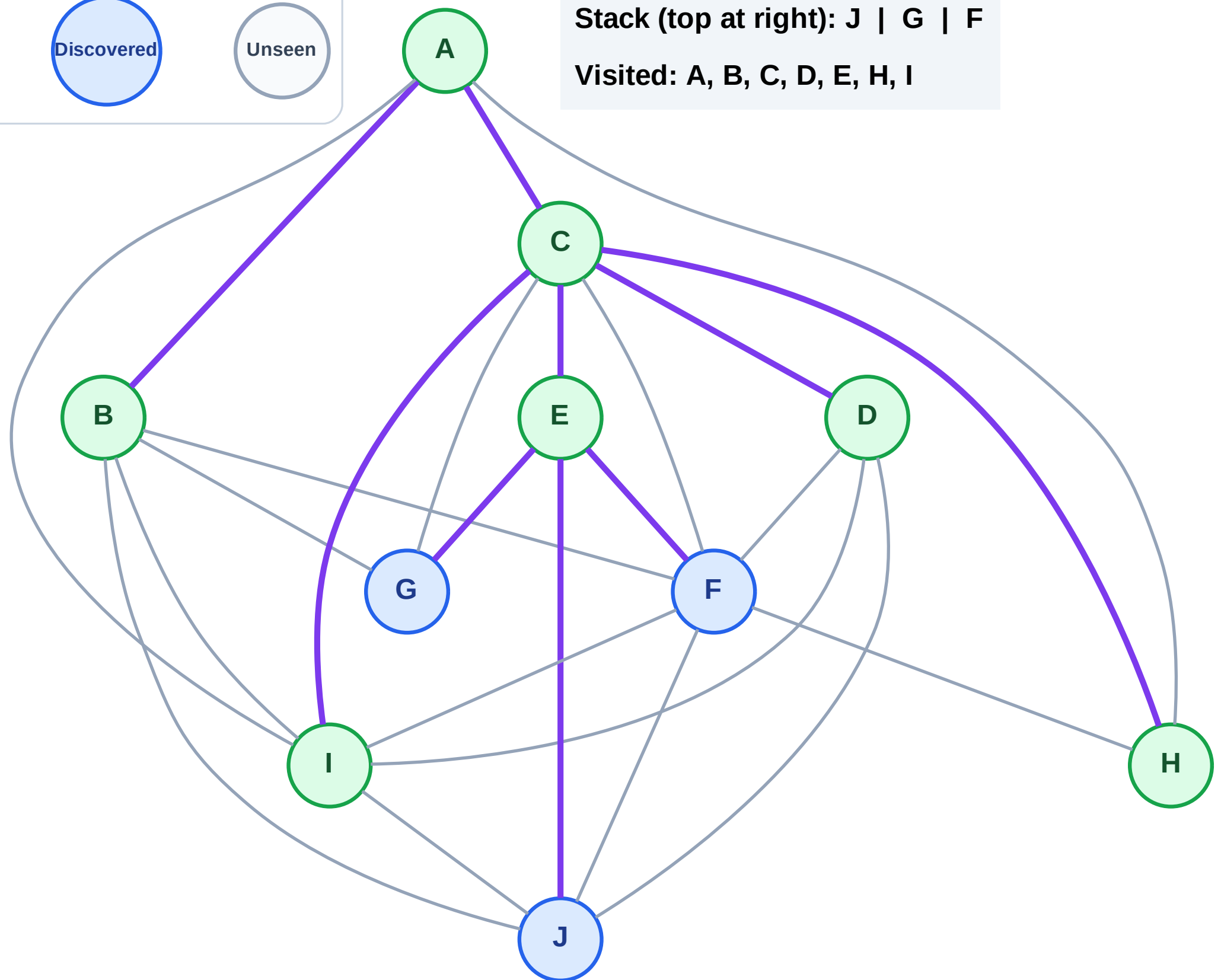
Step 15: No new neighbors from I

Legend



Stack (top at right): J | G | F

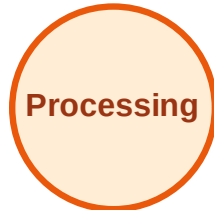
Visited: A, B, C, D, E, H, I



DFS Traversal

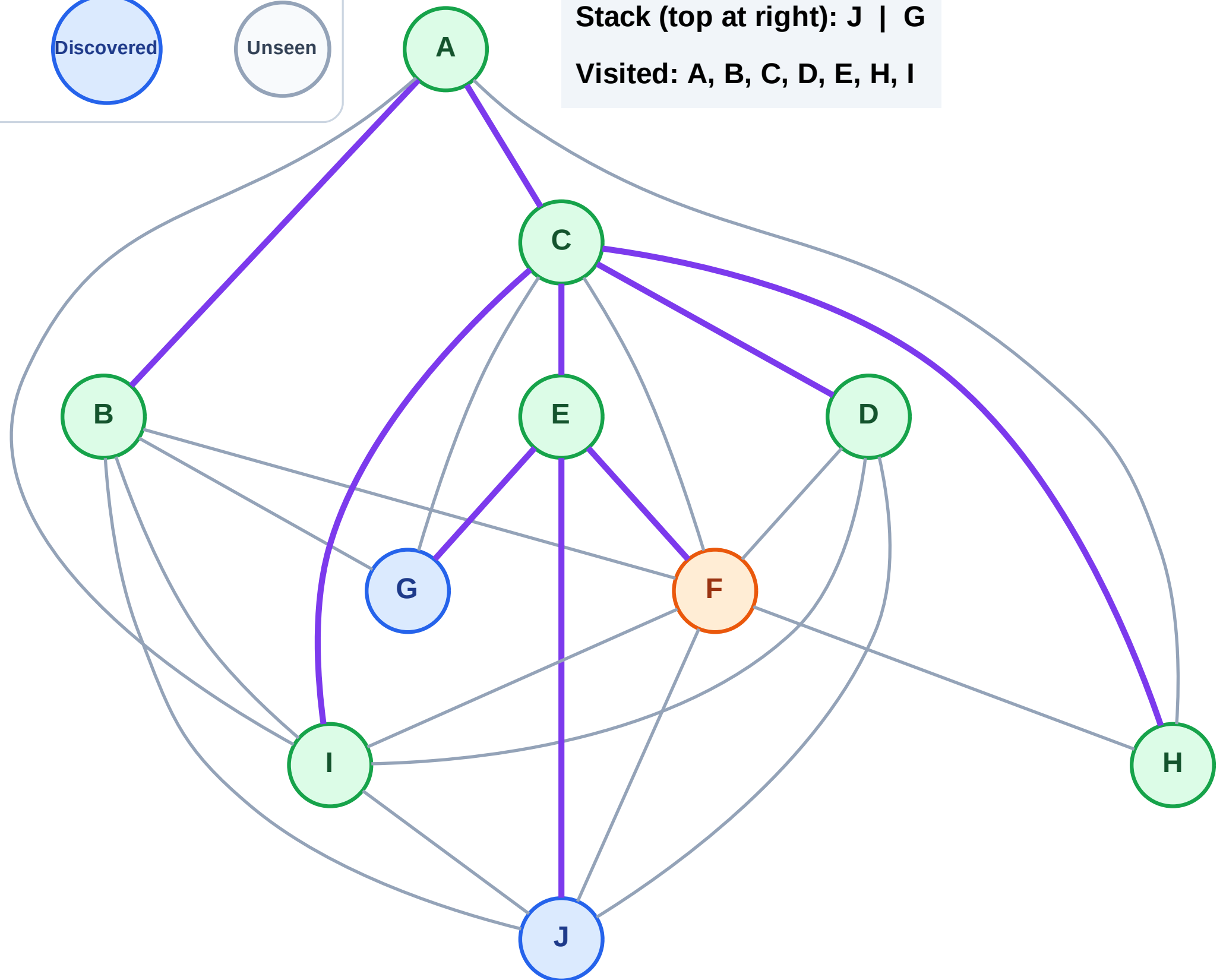
Step 16: Process node F

Legend



Stack (top at right): J | G

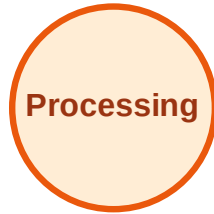
Visited: A, B, C, D, E, H, I



DFS Traversal

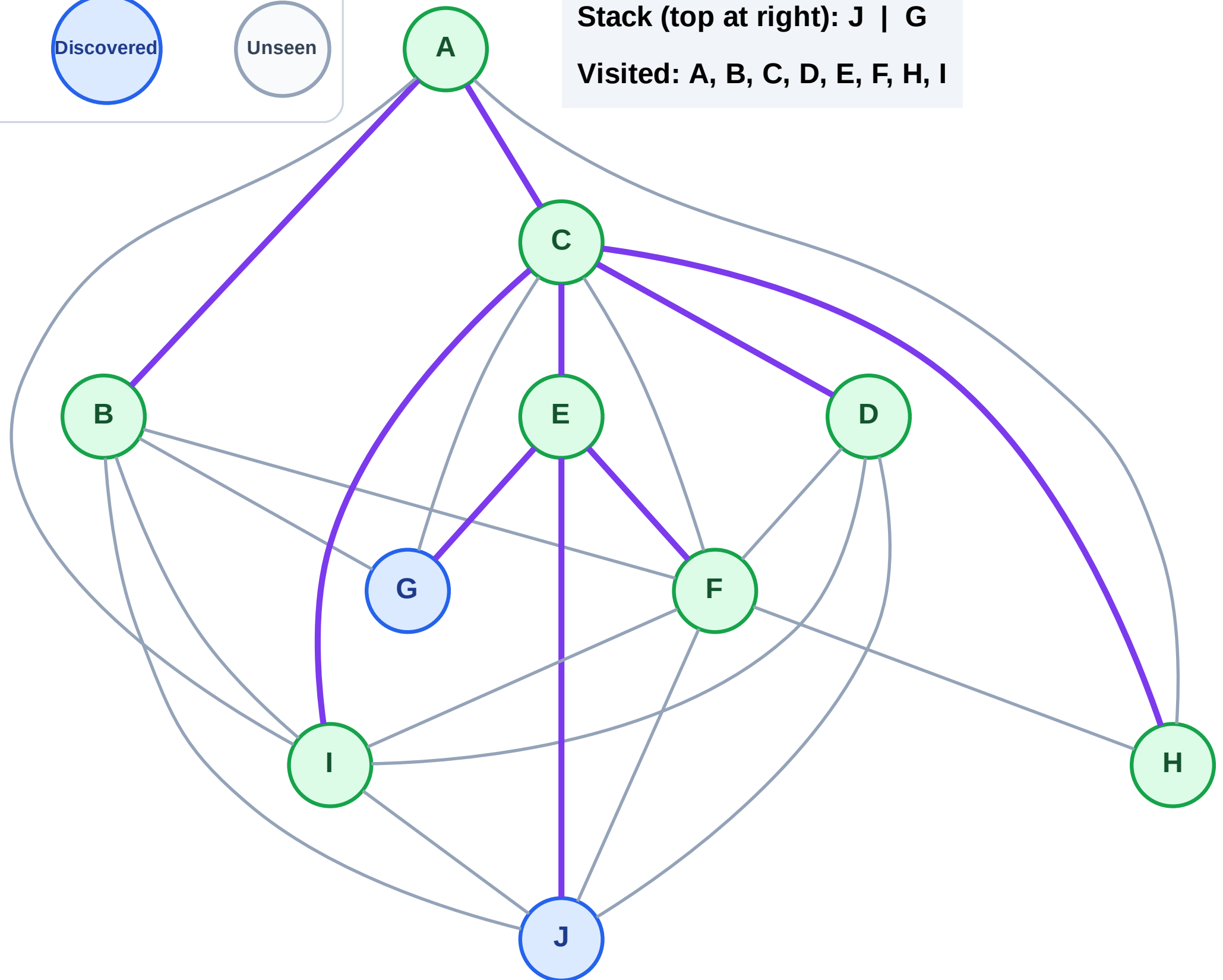
Step 17: No new neighbors from F

Legend



Stack (top at right): J | G

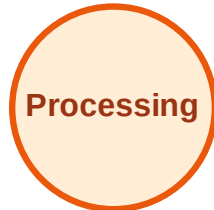
Visited: A, B, C, D, E, F, H, I



DFS Traversal

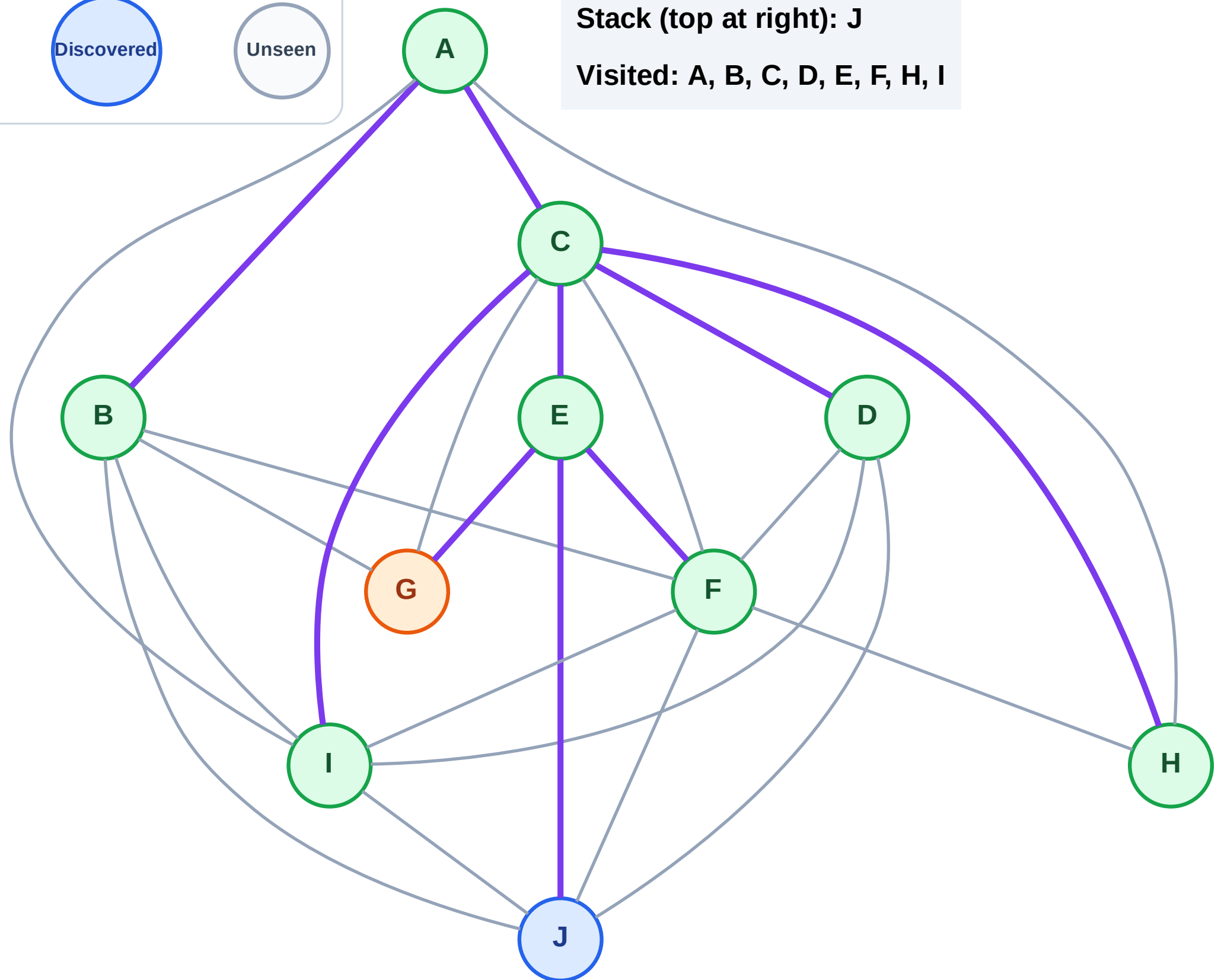
Step 18: Process node G

Legend



Stack (top at right): J

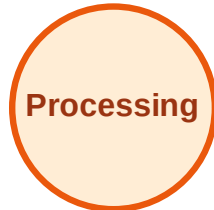
Visited: A, B, C, D, E, F, H, I



DFS Traversal

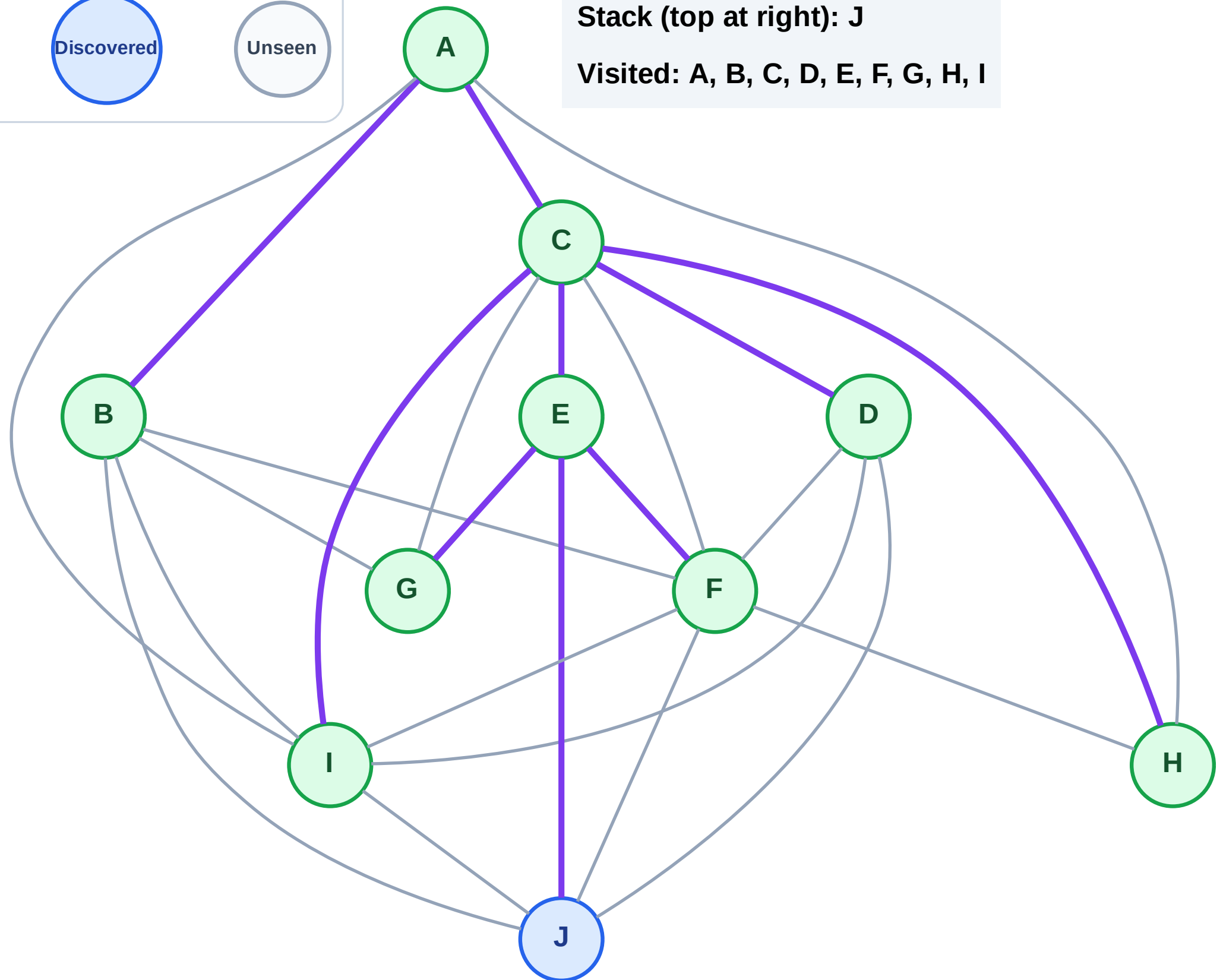
Step 19: No new neighbors from G

Legend



Stack (top at right): J

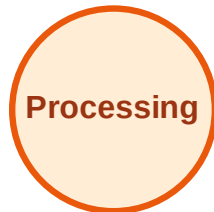
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

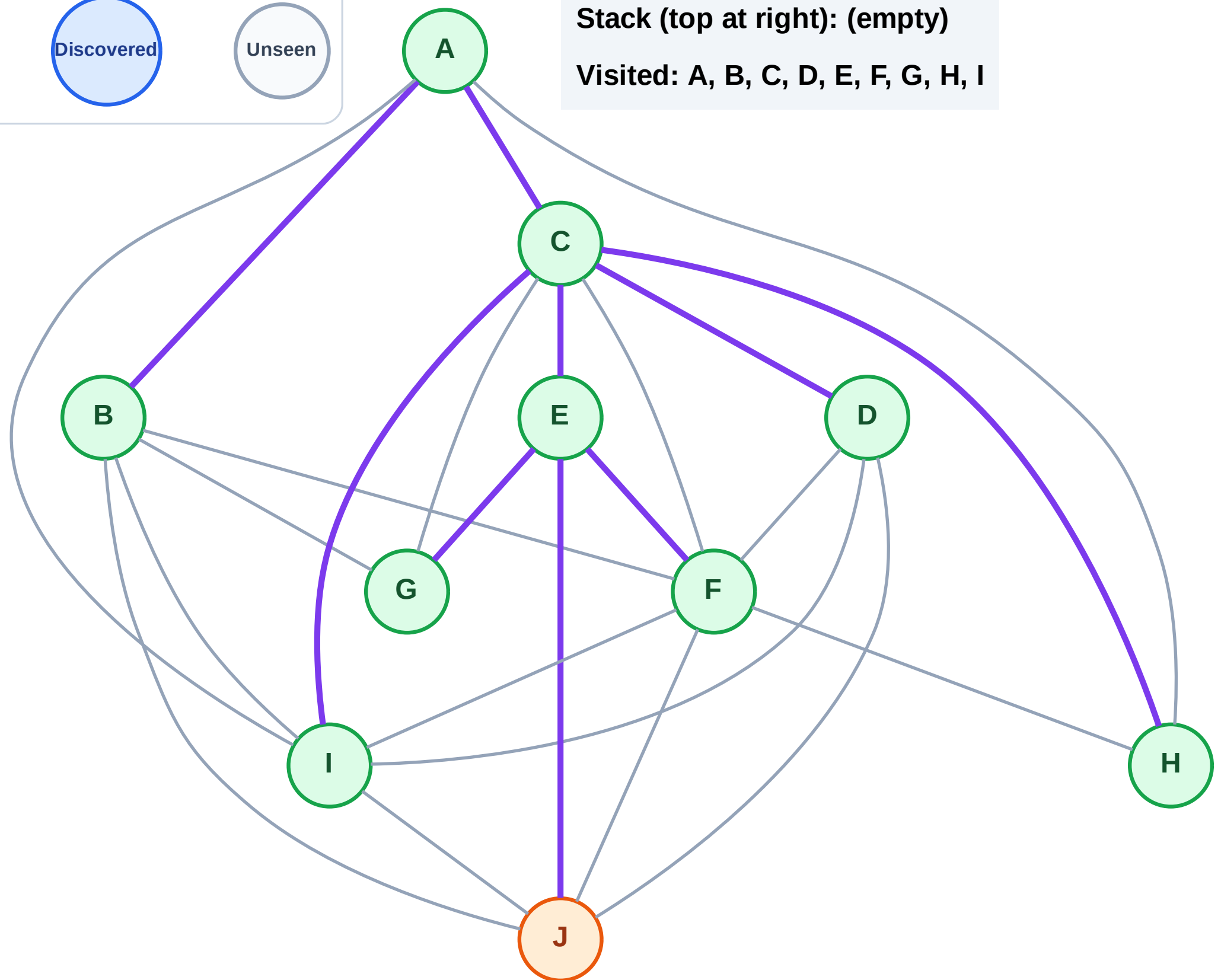
Step 20: Process node J

Legend



Stack (top at right): (empty)

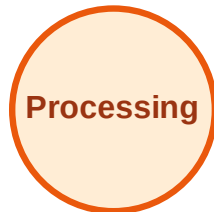
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

